

An Auditable, Calibrated, and Explainable Multi-Agent System for Job-Candidate Recommendation

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Abstract

Online hiring platforms rank millions of resume–job pairs, yet return a single opaque score with no breakdown of which fields drove the rank, so recruiters cannot justify shortlists and candidates cannot act on what would change theirs. We present JOBMATCH, an auditable, calibrated, and explainable recommendation methodology for recruitment (implemented as a multi-agent system) that combines four contributions. First, a composite score that decomposes into six documented channels (semantic similarity, skill overlap, title fit, experience tier, compensation fit, remote policy), giving component-level explanations without post-hoc attribution; alongside it, a graded relation-aware skill matcher that, on a de-circularized benchmark, collapses orthographic/synonym variants to exact (recall 1.00) yet keeps most non-equivalent look-alikes distinct (seven of eight, with one flagged catalog-data merge). Second, a calibrated-confidence layer reported with an honest calibration–discrimination trade-off: Platt scaling attains a low equal-width ECE (0.019) but near-base-rate discrimination (adaptive ECE 0.084, AUC 0.76), whereas beta calibration preserves discrimination (adaptive ECE 0.009, AUC 0.96); we keep Platt as the frozen default and recommend beta. Third, structural *and* mechanistic explanation-faithfulness checks (the latter: does removing the top-attributed channel displace the top-ranked item), plus a 50-pair counterfactual probe. Fourth, a reproducible, regression-gated open-source artifact (frozen 30-resume/15-job/47-label corpus, one-command evaluation). Evaluation is deliberately controlled but small (30 resumes, 15 jobs, 47 human-labeled pairs, single annotator, positive-only labels), so results establish methodological *feasibility*, not population-level ranking superiority: the composite reaches $\text{nDCG@5} = 0.949$ but its gain over the semantic baseline (0.878) is not significant ($p = 0.10$; nothing survives Holm correction at $n = 30$), and a 25-configuration protocol-gated search finds no better ranker. Decomposing the skill channel, an asymmetric coverage form drives the synthetic gain (Jaccard $0.917 \rightarrow 0.949$) while the relation-aware credit — the novel component — helps only on the real corpus (six of thirty queries improve, none worsen; sign-test $p = 0.03$), which we report as a directional signal rather than a broad effect.

Keywords: Recommender systems, Explainable AI, Confidence calibration, Skill matching, Job-candidate matching, Auditable ranking, Recruitment

1. Introduction

Online hiring platforms process millions of resume–job pairs annually, and the systems that rank them are increasingly load-bearing for both the candidate and the recruiter (Tang et al., 2025). A typical applicant tracking system (ATS) receives a resume, parses it into a structured representation, scores the resume against an open job requisition, and returns a ranked shortlist. The scoring function is, in the dominant production paradigm, a single number with no breakdown of which resume fields or job requirements drove the rank (Koren et al., 2009; He et al., 2017). This opacity is not an oversight; it is the cost of optimizing for offline ranking metrics at scale, and the cost is paid by the people whose careers depend on the output.

1.1. The engineering problem

The recruitment domain has a documented accountability gap that follows directly from opaque ranking. A recruiter who has shortlisted 20 candidates from a pool of 200 cannot, with the current generation of ATS tools, point to the specific job requirements that moved a candidate into or out of the shortlist. A candidate who has edited a resume to add a new skill has no way to see whether the new skill moved a posting up or down the list, or whether the change reached the system at all. A compliance officer who needs to audit a hiring decision for disparate-impact liability under equal-employment law (Raub, 2018; Ajunwa, 2021) has no access to the per-decision factor decomposition that would make the audit tractable. The accountability gap is an engineering problem with a known

cost: time-to-hire rises when recruiters must verify shortlist decisions manually, candidate satisfaction drops when applicants cannot act on opaque rankings, and the regulatory exposure of the platform grows when the scoring function is not auditable.

1.2. Regulatory and engineering context

The accountability gap is not hypothetical. Recent regulatory frameworks in the European Union, the United Kingdom, and several US jurisdictions have begun to require auditable explanations of algorithmic decisions that affect employment outcomes (Raub, 2018; Ajunwa, 2021). This growing scrutiny makes audibility and per-decision documentation increasingly important for algorithmic systems that affect employment decisions; and the cost of retrofitting an auditable explanation layer onto a deployed opaque system is typically higher than the cost of designing for audibility from the start. We do not claim compliance with any specific statutory provision — establishing that is a legal question beyond this paper’s scope — only that decomposable, documented decisions are the engineering prerequisite for such audits. The engineering cost of opaque ranking is also concrete: verifying shortlist decisions manually consumes a substantial share of recruiter time, and candidates on platforms that expose no per-decision feedback have no actionable signal for how to improve their standing (we state these as qualitative motivations, not measured effects). These costs are the motivation for the present work: an applied AI system that integrates the per-decision factor decomposition, the calibrated confidence, and the explanation generation into a single inference, so that the audibility

and the user-facing transparency are properties of the inference itself rather than properties of a post-hoc attribution step.

1.3. Limitations of existing AI approaches

Existing AI approaches to the recruitment ranking problem fall into three families, each with a known limitation. *Learned-to-rank systems* (He et al., 2017; Covington et al., 2016; Koren et al., 2009) optimize for offline metrics (nDCG, MAP, MRR) on historical interaction data and return a single score. They are accurate on aggregate metrics but opaque on individual decisions, and they are not auditable without a separate attribution step. *Neural semantic matchers* (Reimers and Gurevych, 2019) encode resumes and job descriptions into a shared embedding space and return a similarity score. They are flexible and language-aware but treat the score as a black-box number and do not decompose the score into interpretable factors. *LLM-based systems* (Wei et al., 2022; Yao et al., 2023; Sumers et al., 2024) are increasingly used for resume parsing, job description normalization, and explanation generation, but they are not yet used as the primary ranker in production systems; their latency, cost, and non-determinism are blocking issues for the hot path of a high-throughput ATS.

1.4. Why current approaches are insufficient

The three families share a common limitation: none of them returns, in a single inference, a ranked list with both per-decision factor decomposition and a calibrated confidence value. The factor decomposition is what makes an explanation auditable; the calibrated confidence is what makes a

confidence display trustworthy. A ranking system that returns only a single number requires a separate attribution step (which is post-hoc, not part of the inference) and a separate calibration step (which is not part of the inference either), and the two steps may not be consistent. A ranking system that integrates the factor decomposition and the calibrated confidence into the inference itself is a more reliable engineering artifact, and the engineering benefit compounds when the inference also includes an explanation that is anchored to the same factors.

1.5. Research gap

The research gap is a lack of applied AI systems for the recruitment domain that integrate three properties in a single inference: a multi-channel composite ranking with factor decomposition, a calibrated confidence value, and a component-level explanation that is bound to the channels that produced the score. Existing systems address one or two of the properties but not all three. A production-grade recruitment ranking system requires all three because the recruitment workflow is a multi-stakeholder decision process (candidate + recruiter + compliance) in which each stakeholder requires a different view of the same ranking.

1.6. Our contribution

We address this gap with JOBMATCH, an auditable, calibrated, and explainable recommendation methodology for the recruitment domain, implemented as a modular multi-agent system (the agent decomposition is an engineering/failure-isolation choice, not the scientific contribution; see

Section 5.8). The architecture decomposes the ranking workflow into a small set of role-separated components, each with a single user constituency and a clearly defined interface to the others. The contribution is fourfold: (i) A composite ranking with six explicit channels (semantic similarity, skill overlap, title fit, experience tier, compensation fit, remote policy), each contributing a documented weight, enabling component-level explanations without post-hoc attribution; and, for the skill channel, a graded relation-aware matcher (exact/related/unrelated with partial, never full, credit for related skills) defined in Section 3 and evaluated in Section 5.8 on a de-circularized benchmark. (ii) A calibrated confidence display that applies Platt scaling (Platt, 1999; Guo et al., 2017) to the composite ranking output and reports a confidence value alongside the ranked list, reducing the held-out 5-fold expected calibration error to 0.019, while we report that the calibrated confidence has limited discrimination (a stated limitation). (iii) An automated, component-level faithfulness evaluation suite (structural checks plus a mechanistic ranking-level probe, not a human faithfulness study) that quantifies whether explanation bullets correspond to the channels that produced the score, and a counterfactual probe that tests whether single-field edits predicted by the explanation move the rank. (iv) A reproducible engineering surface: the prototype, the frozen demo corpus (30 resumes, 15 jobs, 47 labeled pairs), the explanation generator, the calibration layer, and a 341-test regression-gated benchmark are released as an open-source artifact.

1.7. Application context

Figure 1 shows the prototype candidate-side portal in operation: the candidate has uploaded a resume, the system has parsed and normalized the profile, and the ranked list of jobs is returned with the per-decision factor decomposition (semantic fit, skills overlap, experience fit, compensation fit, remote fit), the calibrated confidence, and the matched and missing skill tokens visible inline. Figure 2 shows the symmetric prototype employer-side portal: the recruiter has pasted a job description, the system has parsed the requirements, and the ranked list of candidates is returned with the same factor decomposition and a per-candidate explanation drawer. Figure 3 gives a comprehensive view of the same portal surface, covering all eight primary interaction states in the order a user encounters them: candidate-side (a) resume upload, (b) parsed-fields confirmation, (c) match list with explanation, and (d) counterfactual view; employer-side (e) job upload, (f) requirements confirmation, (g) reverse match list, and (h) shortlist panel. A candidate or recruiter interacts with a transparent interface that surfaces the per-decision factor decomposition and the calibrated confidence alongside the ranked list. The system is a research prototype rather than a production deployment; the application consequence is the engineering pattern, not a specific commercial product. Figure 4 shows the high-level architecture; Figure 5 and Figure 6 expand the candidate- and employer-side components; Figure 10 adds the admin/evaluation console and the data plane.

Table 1: The contribution at a glance: each claimed contribution, the mechanism that realises it, and how it is evaluated in this paper. The novelty is the auditable *integration* and its property-by-property evaluation, not a new ranking algorithm.

Contribution	Mechanism	Evaluation
Decomposable ranking	six explicit weighted channels (fixed priors)	nDCG@5 + leave-one-channel ablation (Section 5.1)
Relation-aware skills	exact / related / unrelated graded credit	de-circularized skill benchmark + decomposition (Section 5.2)
Calibrated confidence	Platt / beta / isotonic on the composite score	ECE + adaptive ECE + Brier skill + AUC (Section 5.3)
Explanation faithfulness	score-bound templates + mechanistic probe	structural checks + rank-displacement probe (Section 5.4)
Counterfactual robustness	controlled single-field profile edits	rank / score sensitivity (Section 5.5)
Reproducibility	frozen artifacts + one-command CI gate	341 regression tests, verifier-gated numbers (Section 5.6)

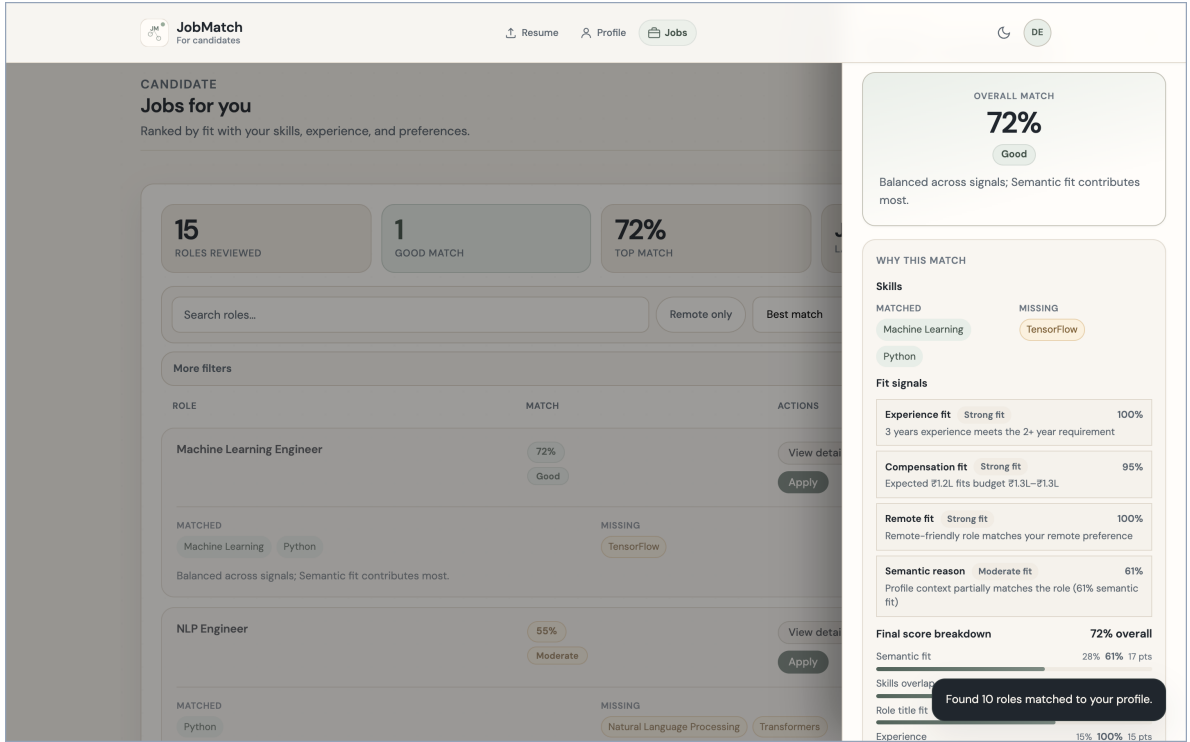


Figure 1: Prototype candidate-side portal: ranked job list with the per-decision factor decomposition (skills overlap, experience, compensation, remote, semantic) and the calibrated confidence (72%, "Good") shown both as a header and as an expandable drawer on the right of the selected job. Matched skill tokens (Machine Learning, Python) and missing tokens (TensorFlow) are surfaced inline so the candidate can act on the output without leaving the page.

1.8. Paper organization

The remainder of the paper is organized as follows. Section 2 reviews the literature on AI-based

recommendation, semantic matching, knowledge-driven AI, LLM agents, explainable AI, and trustworthy AI. Section 3 describes the proposed

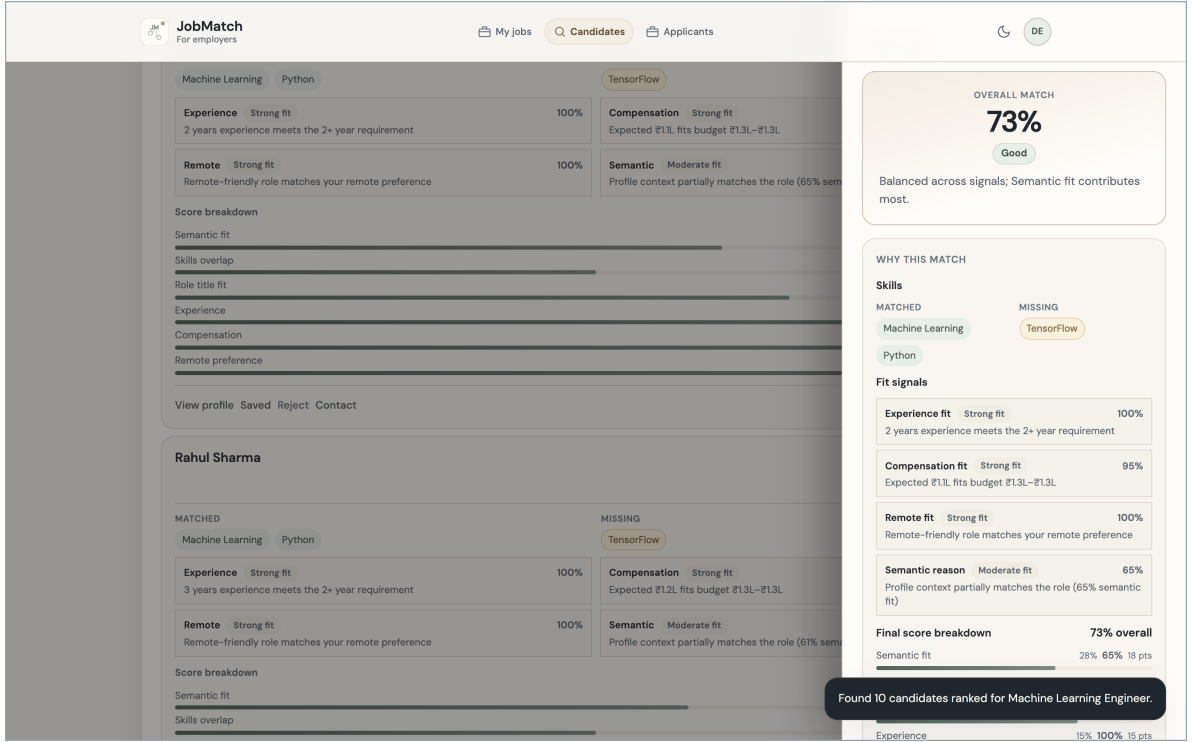


Figure 2: Prototype employer-side portal (reverse direction): ranked candidate list for a posted job, with the per-candidate factor decomposition (semantic, skills, experience, compensation, remote), the per-candidate calibrated confidence, and the inline action menu (view profile, saved, reject, contact). The recruiter sees the same explanation structure that the candidate sees; the scoring pipeline is symmetric and the only difference is the direction of the query.

methodology, including the multi-agent architecture, the knowledge representation, the composite ranking, the explanation generation, the confidence calibration, and the implementation details. Section 4 describes the experimental setup, including the corpus, the baselines, the metrics, and the evaluation protocol. Section 5 reports the results, including the main ranking comparison, the faithfulness evaluation, the calibration results, the counterfactual probe, and the ablation study. Section 6 discusses the engineering implications and the limitations of the evaluation. Section 7 outlines the future work. Section 8 concludes.

2. Related Work

The work reported here sits at the intersection of six research streams: AI-based recommendation systems, semantic matching models, knowledge-driven AI, LLM-based agents, explainable AI, and trustworthy AI. We briefly review each stream and identify the gap that motivates our system.

2.1. AI-based recommendation systems

Modern recommendation systems rely on learned representations, often combined with sparse lexical signals, to produce personalized rankings. Matrix factorization (Koren et al., 2009) was the dominant

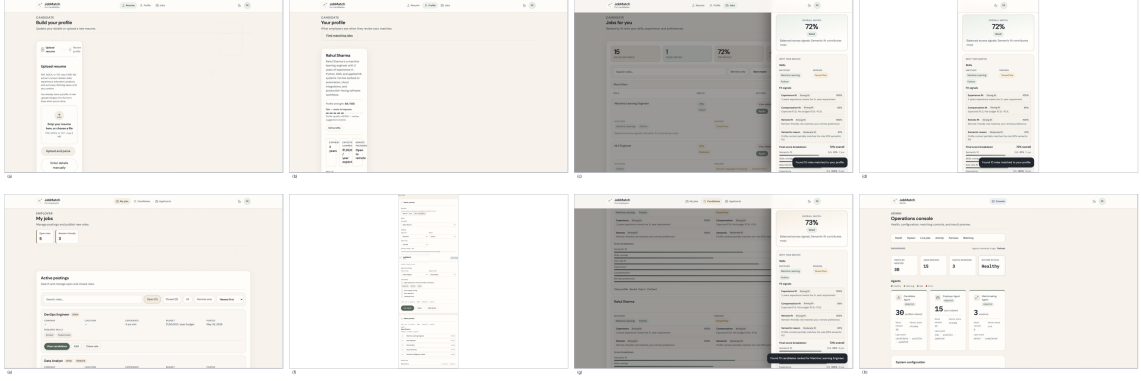


Figure 3: Comprehensive view of the prototype portal surface, showing all eight primary interaction states in the order a user encounters them. Top row (candidate side): (a) resume upload and review; (b) parsed-fields confirmation; (c) match list with component-level explanation drawer; (d) counterfactual view showing how a small edit to a snapshot field would move the rank. Bottom row (employer side): (e) job-description upload and review; (f) parsed-requirements confirmation; (g) reverse match list with component-level explanation; (h) shortlist panel. The admin/evaluation console (event strip, agent health, manual match, demo reset) is not shown here. The single-state views of Figs. 1 and 2 correspond to panels (c) and (g) of this figure.

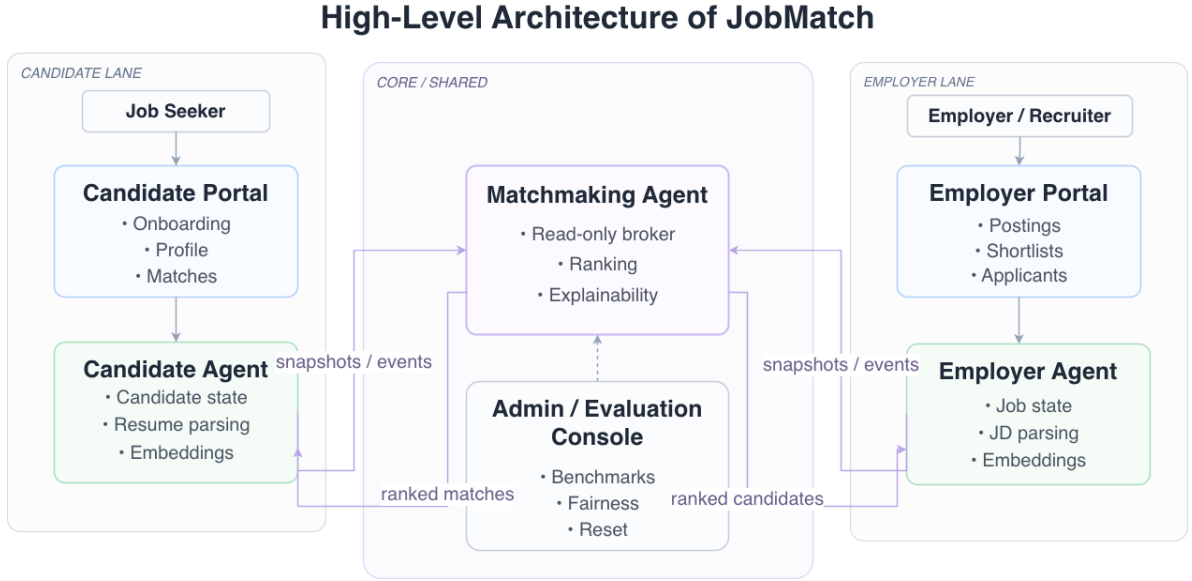


Figure 4: High-level layout of JOBMATCH. The candidate-side component (C1) and the employer-side component (C2) own the user-editable state on each side of the market. The matchmaking component (C3) reads those snapshots, returns a ranked list with component-level reasons and a calibrated confidence, and never writes into either store. The dashed arrows represent snapshot and event flow; the solid arrows represent user-facing recommendations and feedback. The portals shown above (Figs. 1 and 2) are two of the three role-specific portals (candidate, employer, administrator) sharing the same scoring and explanation pipeline.

approach in the 2010s; neural collaborative filtering (He et al., 2017) and deep retrieval (Covington et al., 2016; Le et al., 2020) replaced it in the late 2010s; sequential models (Hidasi et al., 2015) added temporal context in the early 2020s. The dominant production paradigm is a two-tower neural model: one tower encodes the candidate (user, query, resume), the other encodes the job (item, document), and a similarity score is computed in the shared embedding space. The two-tower paradigm is accurate and scalable but opaque: the similarity score is a single number, and the per-decision factor decomposition is not part of the inference. In the recruitment domain specifically, recent ESWA-published work (Rosenberger et al., 2025; Saito and Sugiyama, 2024) has shown that the shared-embedding-space approach (CareerBERT, ESCO-aligned) and the multi-behavior preference modeling approach (Saito & Sugiyama) both reach strong ranking quality on real-world job-corpora, but neither produces a per-decision factor decomposition alongside the score; the present work closes that gap. A 2025 systematic review of 85 person–job recommendation studies (Tang et al., 2025) identifies explainability as a primary open challenge and classifies existing methods into data, model, and output layers; the present work contributes to the output layer (component-level explanations) and the data layer (knowledge-grounded retrieval). Our system retains the two-tower paradigm as one of six channels but adds an explicit composite score that includes non-embedding channels (skill overlap, experience tier, compensation, remote policy), enabling a per-decision factor decomposition that the two-tower paradigm does not provide.

2.2. Semantic matching models

Semantic matching is the retrieval-and-ranking core of any system that compares two documents. Lexical baselines (BM25, TF-IDF) (Robertson et al., 1995) remain competitive on small document pools because they are sensitive to exact term matches; dense semantic matchers (Sentence-BERT) (Reimers and Gurevych, 2019) generalize across paraphrases but lose the term-match sensitivity; hybrid models fuse both signals and are the strongest in the published benchmarks (Cormack et al., 2009). Cross-encoder rerankers are the strongest single model on most benchmarks but are too slow for the hot path of a high-throughput system. Our system uses BM25 and Sentence-BERT as two of the six channels, fuses them with a learned reciprocal rank fusion, and reports a cross-encoder result in the offline evaluation for completeness (with the cross-encoder disabled by default in the prototype ranking path).

2.3. Knowledge-driven AI systems

Knowledge-driven AI systems use a structured representation of domain knowledge (an ontology, a knowledge graph, a vocabulary) to constrain or augment the inference (Nikolakopoulos et al., 2023; Gentile et al., 2014). In the recruitment domain, the structured representation is typically a skill vocabulary (a normalized set of skill names and aliases), an experience taxonomy (entry / mid / senior / staff), and a job-level classification (full-time / part-time / contract / internship). Standardized competency ontologies such as ESCO (European Commission, 2017) and O*NET (U.S. Department

of Labor, Employment and Training Administration, 2024) provide large, curated skill hierarchies, and taxonomy- or ontology-driven skill matching is well established in person–job matching. Our graded matcher is deliberately a *small, hand-built grouping* in this tradition rather than an alignment to a full standard: it is not a new matching mechanism but an auditable, decomposable instantiation whose relation classes and credits are transparent and frozen; we discuss its coarseness and coverage as limitations in Section 6. Our system uses a skill vocabulary as the shared representation across the candidate-side and employer-side components; the vocabulary is consulted by both components and is the only shared state that crosses the role boundary. The benefit of the vocabulary is that the matching layer can compare two documents on the basis of normalized skill tokens rather than free-text skill descriptions, which reduces the failure mode of two resumes that describe the same skill differently.

2.4. LLM-based agents

LLM-based agents decompose complex tasks into tool calls, intermediate reasoning steps, and communication among specialist components (Wei et al., 2022; Yao et al., 2023; Sumers et al., 2024). In the recruitment domain, LLM-based agents are increasingly used for resume parsing (extract structured fields from a free-text resume), job description normalization (extract structured requirements from a free-text posting), and explanation generation (produce a natural-language rationale for a ranking decision). Recent multi-agent designs for the recruitment domain (Lo et al., 2025; Bhattacharya and Verbert, 2025) have demonstrated

that LLM-driven agentic frameworks can match or approach human-rater performance on resume screening (Lo et al., 2025, on a hybrid RAG-LLM framework; Bhattacharya & Verbert, 2025, on the xCUI conversational system with a 20-participant user study). Our system uses an LLM for parsing and for explanation generation but does not use the LLM as the primary ranker; the LLM is on the cold path (parsing) and on the post-rank path (explanation), not on the hot path (ranking). The decision reflects a deliberate engineering trade-off: the hot path must be deterministic, fast, and auditable, and current LLM technology does not meet all three requirements at production scale.

2.5. Explainable AI for recommendation

Explainable AI for recommendation is a well-developed research stream with both post-hoc and intrinsic approaches (Miller, 2019; Gunning et al., 2019; Tintarev and Masthoff, 2015; Peake et al., 2018; Chen et al., 2022). *Post-hoc* approaches train a separate explainer model on the outputs of a black-box ranker; the explainer is not part of the inference and may not be consistent with the ranker. *Intrinsic* approaches build the explainer into the ranker; the explanation is part of the inference and is consistent with the ranker by construction; foundational work in this direction for recsys includes Tintarev & Masthoff’s design framework for explanation interfaces (Tintarev and Masthoff, 2015) and Peake & Wang’s explanation-mining approach for latent factor models (Peake et al., 2018). The distinction matters in the recruitment domain because intrinsic approaches provide the engineering prerequisite for an auditable explanation without further

engineering, whereas post-hoc approaches must be re-validated whenever the underlying ranker is updated. In the recruitment domain specifically, Wanner et al. (Wanner et al., 2020) demonstrate that counterfactual explanations can be generated for explainable AI in recruitment, and Löfström et al. (Löfström et al., 2024) introduce a calibrated-explanations framework that adds uncertainty information to rule-based explanations; the present work combines both directions in a single system and extends the calibration to a multi-channel composite rather than a single classifier. Our system uses an intrinsic approach: the rule-based explainer is bound to the same six channels that produce the score, and the explanation bullet that names a channel is guaranteed to correspond to that channel’s contribution to the score. Counterfactual explanations (Kaur et al., 2020; Wanner et al., 2020) are a complementary approach that explains a ranking by identifying the smallest change to the input that would change the output; our system includes a counterfactual probe that validates whether the predicted change is realized in the next ranking. The combination of an intrinsic per-decision factor decomposition, a calibrated confidence display, and a counterfactual probe is, to our knowledge, novel in the recruitment-recommendation literature.

2.6. Trustworthy AI and confidence calibration

Trustworthy AI is the research stream that treats trust as a property of the inference, not of the model’s reputation (Liu et al., 2020; Pu and Chen, 2007). A trustworthy ranking system reports, alongside the ranking, a calibrated confidence value that the user can read as a signal of the system’s

uncertainty. The standard measure of calibration quality is the expected calibration error (ECE) (Guo et al., 2017), and the standard recalibration method is Platt scaling (Platt, 1999) or temperature scaling. The recent calibrated-explanations framework of Löfström et al. (Löfström et al., 2024) provides a principled approach to adding uncertainty information to rule-based explanations; we adopt the same conceptual approach of pairing a calibrated confidence with a factor decomposition, applying standard scalar-score calibration to the composite output. Our system applies Platt scaling to the composite ranking output and reports the calibration parameters (slope and intercept) and the Brier score (Brier, 1950) alongside the ECE. We apply standard Platt scaling to the scalar composite score; because the calibrator operates on the single final score, this is ordinary ranker-score calibration rather than a new calibration mechanism, and we make no novelty claim for the act of calibrating a multi-channel score. Our calibration contribution is instead the honest calibration-versus-discrimination trade-off analysis reported in Section 5 (Platt attains a low equal-width ECE but near-base-rate discrimination, while beta calibration attains the lowest ECE *and* preserves discrimination; see Section 5.4). Fairness in information access (Ekstrand et al., 2022) provides a complementary lens: ranking systems in employment contexts can encode proxy variables that disproportionately disadvantage protected groups, and the engineering remedy is to test for disparate impact under controlled perturbations of the input. Causal and counterfactual debiasing is a complementary direction for reducing disparate impact in recommendation;

our system reports an adversarial fairness probe as an engineering sanity check and a foundation for the larger corpus study described in Section 7. The HCI literature on explainable AI user experiences (Amershi et al., 2019; Liao et al., 2020) provides design guidance for the user-facing explanation interface; the present system follows the same engineering pattern of surfacing a calibrated confidence value and a per-decision factor decomposition in the primary user interface, and reports the user-facing surface in Section 1.7.

2.7. Positioning

The present work is positioned as an applied AI system for the recruitment domain that integrates six research streams into a single, evaluated pipeline. The closest prior systems address one or two of the streams but not all six; the comparison is taken up in Section 6. The contribution is the integration, the engineering surface, and the controlled evaluation; the individual components (BM25, Sentence-BERT, RRF, Platt scaling, rule-based explanation) are well-known, and the novelty is in their composition and in the evaluation methodology that maps each component to a measurable property.

3. Proposed Methodology

This section describes the proposed methodology in detail. We begin with the problem formulation and notation (Section 3.1) and a methodology overview (Section 3.2), then describe the multi-agent architecture (Section 3.3), the knowledge representation and retrieval layer (Section 3.4),

the composite ranking with component decomposition (Section 3.5), the component-level explanation generation (Section 3.6), the confidence calibration (Section 3.7), and the implementation details (Section 3.8). The end-to-end flow from input to output is shown in Figure 7.

3.1. Problem formulation and notation

We begin with a formal statement of the task and a description of the notation used throughout the section. Let $\mathcal{R} = \{r_1, r_2, \dots, r_n\}$ denote a set of resumes and $\mathcal{J} = \{j_1, j_2, \dots, j_m\}$ denote a set of job descriptions. Each resume r_i is parsed into a structured representation $\tilde{r}_i = (S_i, E_i, C_i, P_i)$ where S_i is a multiset of normalized skill tokens, E_i is an experience tier, C_i is a compensation expectation, and P_i is a remote-work preference. Each job j_k is parsed into a structured representation $\tilde{j}_k = (S_k^{\text{req}}, S_k^{\text{pref}}, E_k^{\text{min}}, E_k^{\text{max}}, C_k, P_k, T_k)$ where S_k^{req} and S_k^{pref} are required and preferred skill tokens, E_k^{min} and E_k^{max} are the experience bounds, C_k is the compensation band, P_k is the remote-work policy, and T_k is the job title. The task is to learn a scoring function $f : \mathcal{R} \times \mathcal{J} \rightarrow \mathbb{R}$ that ranks the jobs for a given resume, and to produce alongside the ranking a per-decision factor decomposition and a calibrated confidence value. The factor decomposition is a vector $d(r_i, j_k) \in \mathbb{R}^6$ where the six components correspond to the six channels described in Section 3.5, and the calibrated confidence is a value $c(r_i, j_k) \in [0, 1]$ that the user can read as a signal of the system’s uncertainty.

3.2. Methodology at a glance

The end-to-end methodology is a six-stage pipeline: inputs (resume or job description, user

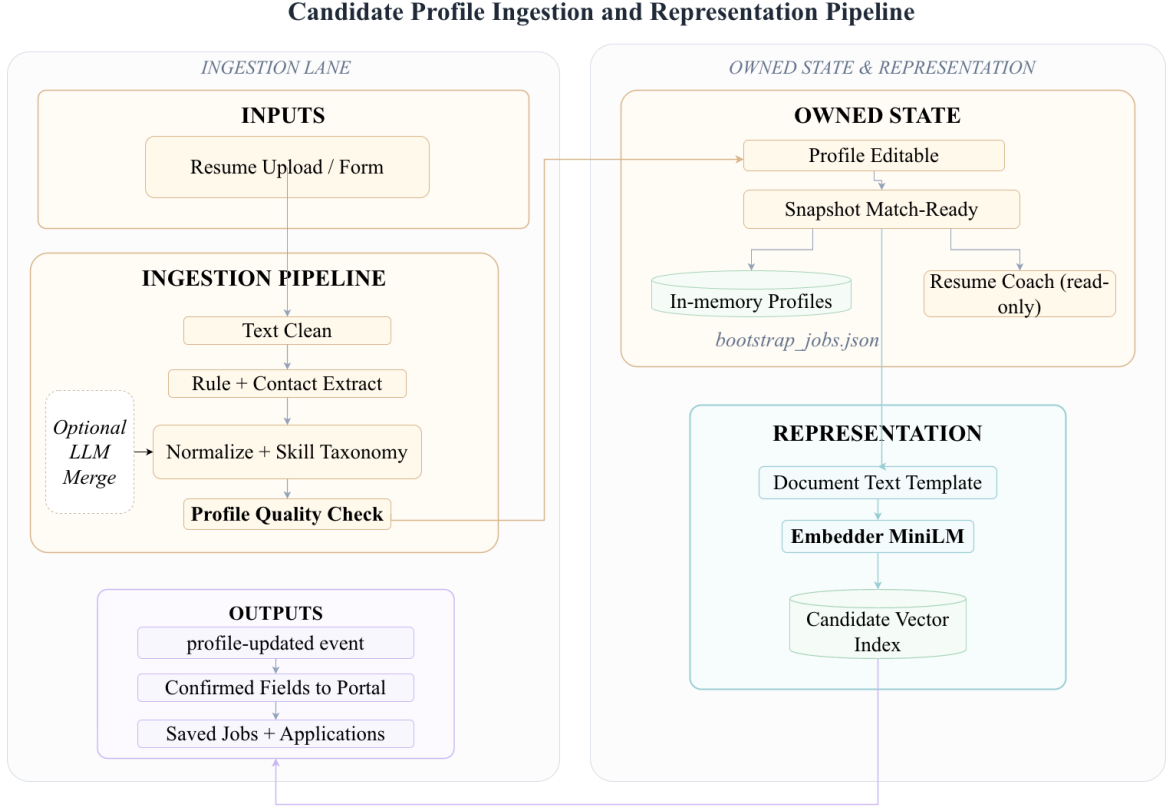


Figure 5: Candidate-side ingestion and representation pipeline. The candidate-side component owns the editable profile and the versioned snapshot. The ingestion lane accepts a resume upload, runs text clean and rule-based extraction, normalizes the result against the shared skill vocabulary (with an optional LLM fallback for ambiguous fields), and presents the parsed result to the user as a review form. Only on user confirmation does the component register a snapshot and emit a profile-updated event.

preferences, and the catalog of normalized skill tokens) flow through knowledge representation, hybrid retrieval, composite ranking, component-level explanation, and confidence calibration before the ranked list, per-channel reasons, and confidence value are returned to the user. Figure 8 visualizes the six stages with the calibrated confidence used only for downstream display alongside the ranked list; the ranking weights themselves are fixed hand-set domain priors, not fitted to the labeled pairs (Section 3.5). The same pipeline is run in both direc-

tions (candidate-initiated and employer-initiated) without modification, and the per-direction user interface is the only stage that is direction-specific. The detailed block architecture, the per-component decomposition, and the per-stage implementation are described in the following subsections.

3.3. Multi-agent system architecture

The architecture decomposes the ranking workflow into three cooperating components, each with a single user constituency and a clearly defined interface to the others. Figure 4 (introduced in Sec-

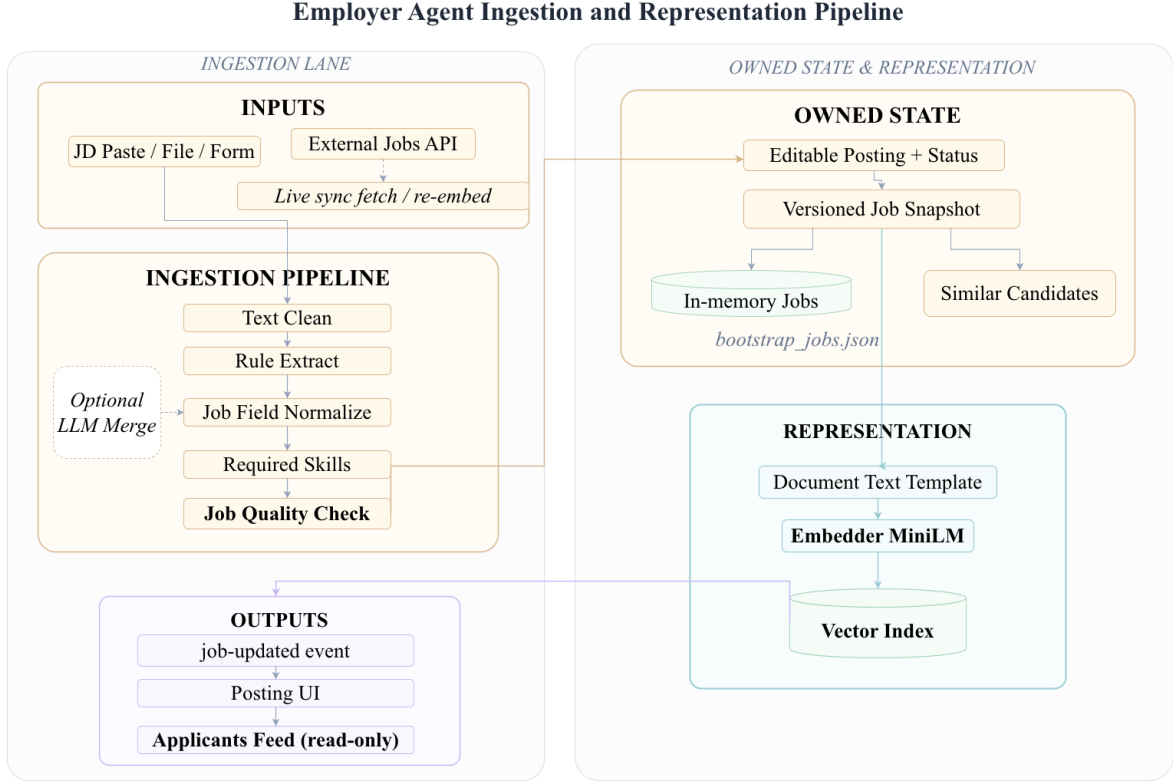


Figure 6: Employer-side ingestion and representation pipeline. The employer-side component mirrors the candidate-side layout for the job side: editable posting, versioned snapshot, and an ingestion lane that accepts a job description paste, file, or external API and presents the parsed result as a recruiter review form. Only on confirmation does the component register a snapshot and emit a job-updated event.

tion 1) shows the high-level layout; Figure 9 below traces the end-to-end matchmaking workflow; Figure 10 gives the detailed system architecture with the admin/evaluation console and the data plane. The role separation is a design-for-accountability choice: the candidate-side component owns the candidate’s editable profile and the candidate’s read-only snapshot; the employer-side component owns the employer’s editable job description and the job’s read-only snapshot; the matchmaking component reads both snapshots, scores each pair, attaches the per-decision factor decomposition, and attaches the calibrated confidence. The role boundary is also the

privacy boundary: the candidate-side component does not see the employer’s editable job description; the employer-side component does not see the candidate’s editable profile; the matchmaking component never writes into either side. The LLM is used by the candidate-side component for resume parsing and by the matchmaking component for explanation generation; the LLM is not on the hot ranking path. The hot path must be deterministic, fast, and auditable, and current LLM technology does not meet all three requirements at production scale; placing the LLM on the cold path is a direct consequence of this constraint. The role-separated

Matching, Ranking and Explanation Pipeline

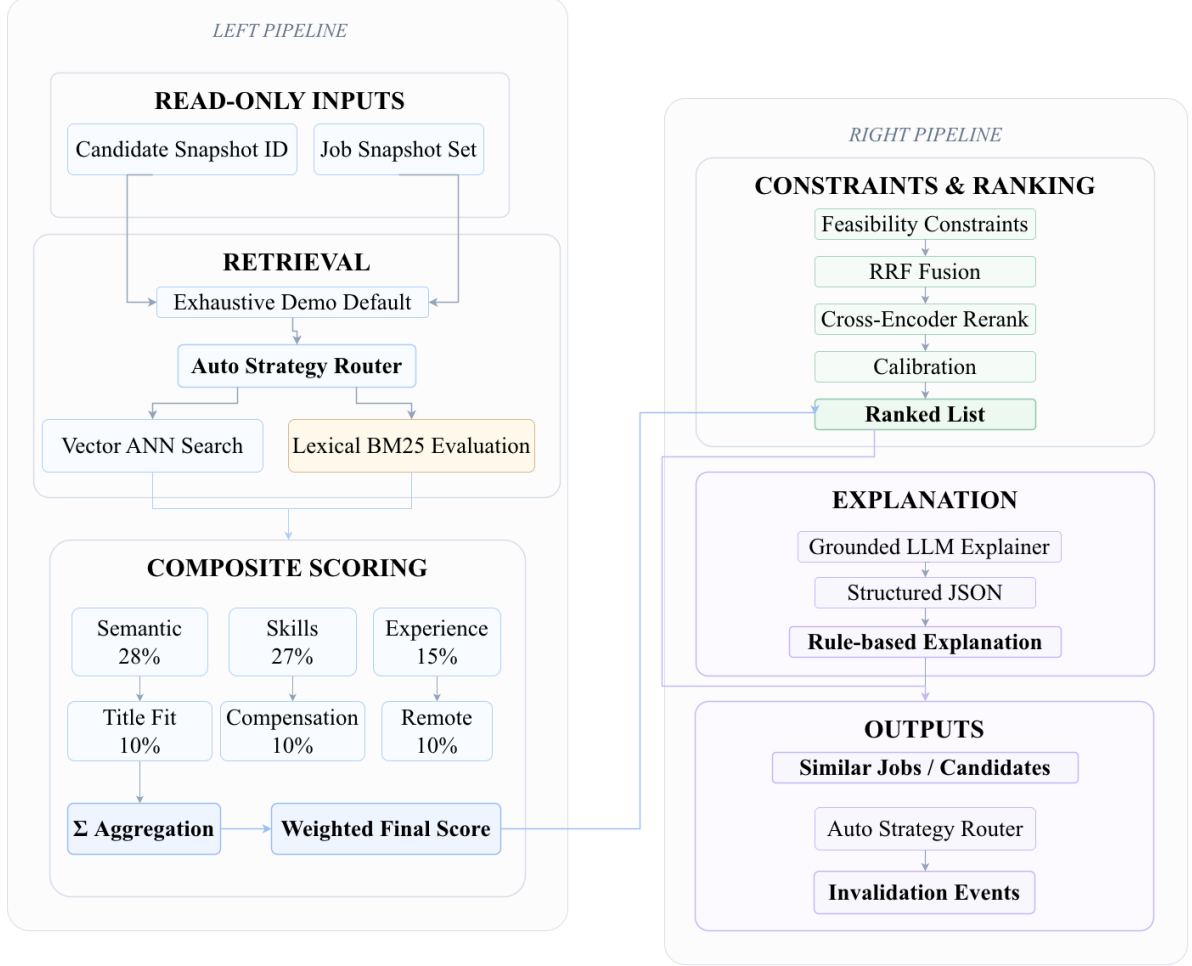


Figure 7: End-to-end methodology: from resume and job-description inputs through knowledge representation, hybrid retrieval, composite scoring, RRF fusion, cross-encoder rerank, calibration, and component-level explanation, to the ranked, explained, and calibrated output. The dashed feedback arrow indicates that the calibrated confidence is recomputed on each ranking request and used to inform downstream display and user actions. The same pipeline runs for both the candidate-initiated and the employer-initiated direction; the matchmaking component does not see the editable profile behind either side.

layout maps to the privacy boundary in production HR systems.

3.4. Knowledge representation and retrieval

The shared knowledge representation is a skill vocabulary $\mathcal{V} = \{v_1, v_2, \dots, v_V\}$ of $V \approx 5,000$ normal-

ized skill tokens with alias mappings from free-text skill descriptions to vocabulary tokens. Both the candidate-side and the employer-side components consult the same vocabulary; a free-text skill description in a resume or job description is mapped to a vocabulary token by a rule-based normalizer

Methodology: from resume/job inputs to ranked, explained, calibrated output

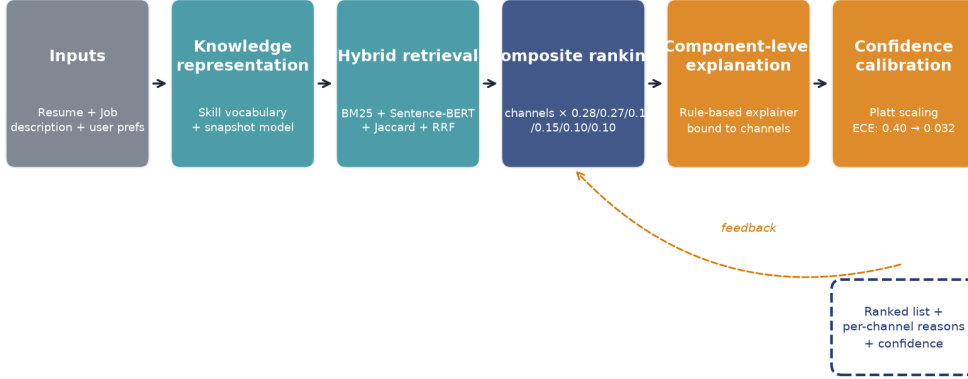


Figure 8: End-to-end methodology: from resume and job-description inputs through knowledge representation, hybrid retrieval, composite scoring, component-level explanation, and Platt-scaled confidence calibration to the ranked list with per-channel reasons and a calibrated confidence value. The dashed orange feedback arrow indicates that the calibrated confidence is recomputed on each ranking request and is available to inform downstream display and user actions. The same pipeline runs in both the candidate-initiated and the employer-initiated direction; the per-direction portal (Figs. 1 and 2) is the only stage that is direction-specific.

with an LLM fallback for ambiguous cases. The retrieval layer is a hybrid of four channels: (i) BM25 (Robertson et al., 1995) over the free-text concatenation of the resume and the job description, with a field-boosted weighting that emphasizes the title field. (ii) Sentence-BERT cosine similarity (Reimers and Gurevych, 2019) over the embedding of the resume’s free-text concatenation and the embedding of the job’s free-text concatenation, using the `all-MiniLM-L6-v2` encoder with $d = 384$. (iii) Jaccard skill overlap over the normalized skill multisets, with a soft-embed weighting that allows partial credit for skill aliases (e.g., "ML" and "machine learning" are mapped to the same vocabulary token but "K8s" and "Kubernetes" are partial aliases). (iv) Reciprocal rank fusion (Cormack et al., 2009)

of the top- K lists from the three channels above, with $K = 10$ and the RRF constant $k = 60$. The four channels are evaluated independently and the results are fused by the composite ranking layer described next. The choice of four channels is motivated by the empirical observation that no single channel dominates across all pairs: BM25 is strong on exact term matches, Sentence-BERT is strong on paraphrases, Jaccard is strong on skill overlap, and RRF is a robust fusion of the three. The hybrid layer returns, for each pair (r_i, j_k) , a four-dimensional vector of channel scores; the composite ranking layer combines the four channels with the two non-retrieval channels (experience, compensation, remote) into a single score.

Employer Lifecycle and Matchmaking Workflow

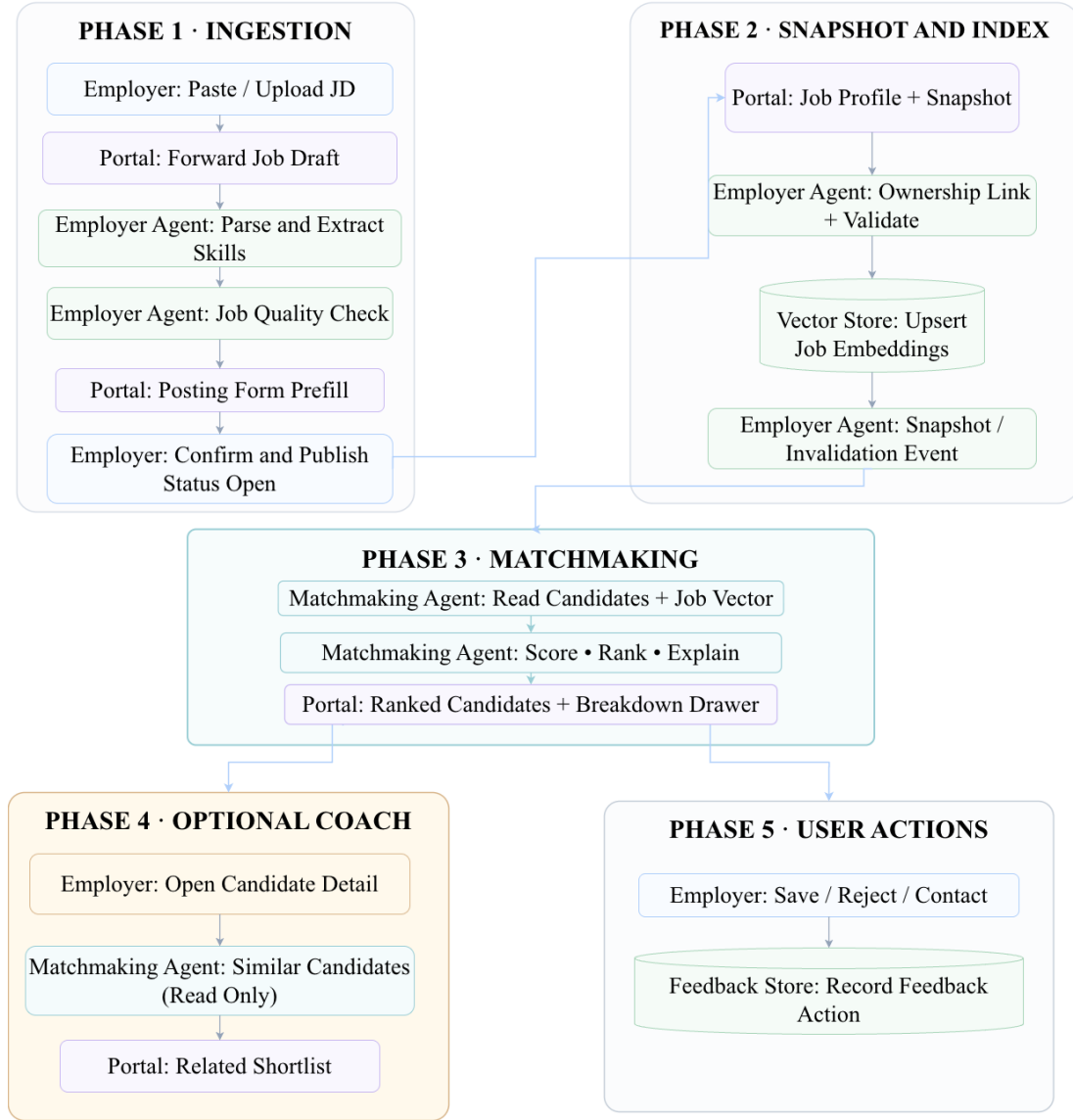


Figure 9: End-to-end matchmaking workflow (employer direction shown; the candidate direction is symmetric). The workflow is the same whether it is initiated by a candidate refreshing their matches or by an employer refreshing their shortlist: ingestion, snapshot/index, ranking, optional coach feedback, and user actions are sequenced so the read-only matchmaking component never crosses the ownership boundary.

3.5. Composite ranking with decomposition

The composite ranking layer combines six channels into a single score. Figure 11 documents the six

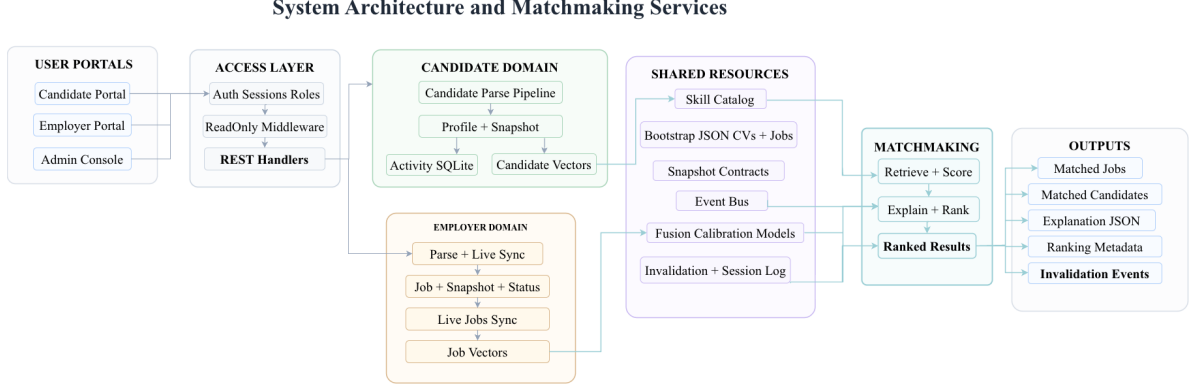


Figure 10: Detailed system architecture and matchmaking services. The user portals (candidate, employer, admin) sit above an access layer with auth, role, and read-only middleware that fronts the REST handlers. The candidate domain owns the parse pipeline, profile, snapshot, and activity log; the employer domain owns the parse + live sync, job + snapshot, and job vectors. The shared resources layer exposes the skill catalog, bootstrap data, snapshot contracts, event bus, fusion and calibration models, and the invalidation log. The matchmaking component reads from the shared resources and from the snapshot edges of each domain, never from the editable stores.

channel weights, and the equation below makes the combination explicit. The two high-weight channels (semantic 0.28 and skill 0.27) carry the dominant share of the score; the four medium- and low-weight channels (title 0.10, experience 0.15, compensation 0.10, remote 0.10) carry the remaining 45%. The weights are *hand-set domain priors*, not fitted to the labeled pairs; a bootstrap weight-stability analysis (Section 5) confirms they behave as a design prior rather than a validation-optimized optimum, and a protocol-gated search over 25 ranking configurations (Section 5) finds no configuration that significantly outperforms this fixed composite on the present corpus.

The six channels are: (1) semantic similarity (the Sentence-BERT cosine from the retrieval layer). (2) skill overlap (the Jaccard from the retrieval layer, with the soft-embed weighting). (3) title fit (a binary indicator of whether the resume’s most recent title matches the job’s title in the same domain,

computed by a rule on the title strings). (4) experience tier fit (a function of the difference between the resume’s experience tier and the job’s experience bounds, with a piecewise-linear penalty for under-qualification and over-qualification). (5) compensation fit (a binary indicator of whether the resume’s compensation expectation is within the job’s compensation band). (6) remote-work preference fit (a binary indicator of whether the resume’s remote-work preference is compatible with the job’s remote-work policy). The six channel scores are combined by a fixed-weight linear combination:

$$\begin{aligned}
 f(r_i, j_k) = & 0.28 s_{\text{sem}} + 0.27 s_{\text{skill}} + 0.10 s_{\text{title}} \\
 & + 0.15 s_{\text{exp}} + 0.10 s_{\text{comp}} + 0.10 s_{\text{remote}}
 \end{aligned} \tag{1}$$

where the weights are hand-set domain priors (not fitted to the labeled pairs) and are reported as part of the system configuration; Section 5 reports a bootstrap stability analysis and a protocol-gated

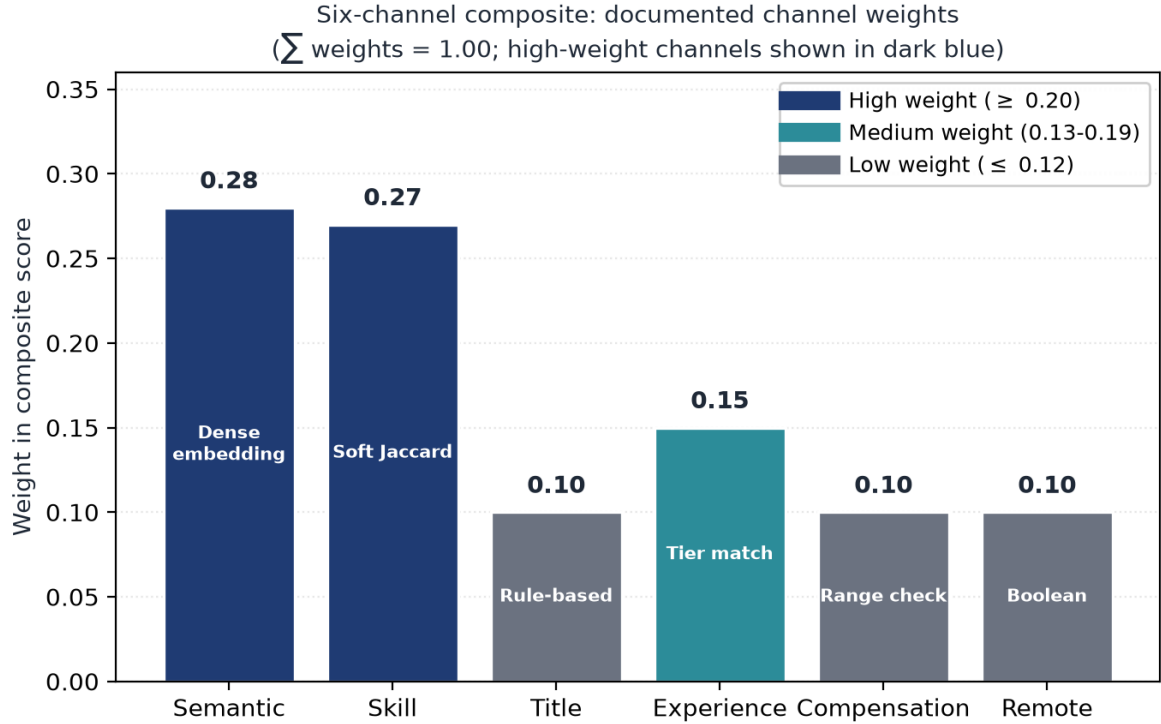


Figure 11: Six-channel composite: documented channel weights. The two high-weight channels (semantic 0.28 and skill 0.27) carry the dominant share of the score; the four medium- and low-weight channels (title 0.10, experience 0.15, compensation 0.10, remote 0.10) carry the remaining 45%. The weights are hand-set domain priors (not fitted to the labeled pairs); see the weight-stability analysis in Section 5.

configuration search that together justify retaining this fixed, auditable prior. The choice of a fixed-weight composite is motivated by two engineering properties: the per-decision factor decomposition is exact (the channel scores are reported in the explanation panel), and the composite is auditable (the weights are documented and the calibration is reported on the same composite).

Relation-aware graded skill matcher. The default skill channel (2) uses binary Jaccard overlap, which gives all-or-nothing credit. As an auditable alternative we define a graded, relation-aware matcher that resolves each (candidate skill, required skill) pair to one of three relation classes

and aggregates by coverage: for each required skill we take the best credit over the candidate’s skills — 1.0 if they share a canonical form after synonym/abbreviation/orthographic normalization (EXACT), 0.5 if they fall in the same skill-taxonomy group (RELATED), and 0 otherwise (UNRELATED) — and average over the required skills. The credits are fixed a priori (not fitted to any evaluation), and RELATED never receives EXACT-level credit. Unlike symmetric Jaccard this form is coverage-oriented (how well the candidate covers the job’s stated requirements) and awards partial, never full, credit to related-but-not-identical skills. Section 5.8 evaluates this matcher on a de-circularized skill-pair

benchmark and, holding all weights fixed, decomposes its ranking effect into the coverage-form change and the relation-aware credit; the primary composite retains binary Jaccard for comparability with the baselines, and the graded matcher is reported as an auditable enhancement rather than adopted as the headline configuration. A learned logistic-regression fusion is reported as an alternative in the offline evaluation; the learned fusion ties the best single configuration but is fit on the same labeled pairs it is judged against, which is an overfitting risk (see Section 6).

3.6. Component-level explanation generation

The component-level explanation is generated by a rule-based explainer that is bound to the same six channels that produce the score. For each channel that contributes more than a configured threshold (default: 5% of the composite score), the explainer emits a bullet that names the channel and the value that drove the contribution. For example, if the skill-overlap channel contributes 0.27 to the composite and the overlap is on the "Python" and "machine learning" skills, the bullet is "Matched on Python and machine learning (skill overlap 0.27)." If the experience-tier channel contributes 0.05 because the resume's experience tier is below the job's lower bound, the bullet is "Below the experience bound by one tier (experience fit 0.05)." The bullets are produced deterministically from the channel scores and the structured representations; the explanation is consistent with the score by construction. An LLM-based explainer is reported as a comparison in the offline evaluation; the LLM-based explainer is a GPT-class model prompted

with the channel scores and the structured representations, and the natural-language rationale is generated by the LLM. The trade-off is that the LLM-based explainer has higher skill-mention coverage (it names a concrete skill in roughly half of the bullets) but lower faithfulness (some bullets are inconsistent with the channel scores because the LLM hallucinates a contribution that the channels did not produce). The rule-based explainer is the default in the prototype; the LLM-based explainer is reported in the offline evaluation as a comparison.

3.7. Confidence calibration

The composite ranking output is a score in $\mathbb{R}$, and the prototype system displays a confidence value in $[0, 1]$ alongside the score. The mapping from score to confidence is learned by Platt scaling (Platt, 1999; Guo et al., 2017; Löfström et al., 2024) on a held-out calibration set of 21 strong and 26 partial labels. The use of Platt scaling for ranking-system calibration follows the calibrated-explanations framework (Löfström et al., 2024), which provides uncertainty information alongside rule-based explanations; here it is applied, in standard form, to the scalar composite score, and we make no novelty claim for the calibration mechanism itself (see Section 2 and the trade-off analysis in Section 5). The Platt scaling parameters are a slope a and an intercept b , applied to the composite score:

$$c(r_i, j_k) = \sigma(a \cdot f(r_i, j_k) + b) \quad (2)$$

where σ is the logistic sigmoid. The calibration parameters are learned by maximum likelihood on the calibration set; the calibration quality is reported as the expected calibration error (ECE), the Brier

score, and a reliability diagram that plots the predicted confidence against the empirical relevance rate over all judged (resume, job) pairs under the held-out 5-fold protocol (Section 4). The uncalibrated composite has $ECE = 0.40$, which is too poorly calibrated to display directly; the calibrated composite has held-out 5-fold $ECE = 0.019$, a large reduction, though the calibrated confidence has low discrimination near the base rate (Section 5.4). The Brier score of the calibrated system is 0.093. The Platt parameters are exposed in the system configuration; a sensitivity analysis on the parameters is reported in Section 5.

3.8. Implementation details

The system is implemented as a small web stack: a Python backend for the parsing, retrieval, ranking, calibration, and explanation logic, and a Node frontend for the candidate, employer, and administrator portals. The system is reproducible from a clean clone: a single command runs the regression benchmark and reproduces the reported numbers within a ± 0.04 nDCG tolerance. The test suite is 302 Python unit tests and 39 Node tests, for a total of 341 tests; the CI gate fails the build on a regression greater than the tolerance. The latency profile is reported in Section 5: the composite bi-encoder completes in approximately 0.4 ms per query, the cross-encoder completes in approximately 141.7 ms per query, and the end-to-end request (including the rule-based explainer) completes in under 10 ms per request. The LLM-based explainer is invoked only when the user requests a natural-language rationale, and is gated by a per-request budget; the typical invocation cost is below the cost of the rank-

ing itself. The prototype, the frozen demo corpus, the synthetic corpus and generator, the evaluation scripts, the explanation generator, the calibration layer, and the regression benchmark are released as an open-source artifact (to be deposited with a citable DOI upon acceptance; see the data availability statement at the end of this paper).

3.9. Engineering surface and integration

The engineering surface of the system is reported here in the level of detail required for a deployment team to reproduce the deployed claims and to integrate the prototype with an existing ATS. The Python backend is structured as a small set of modules: a **parser** module for resume and job description ingestion, a **retrieval** module for the four retrieval channels, a **ranking** module for the composite scoring, an **explanation** module for the rule-based and LLM-based explainers, and a **calibration** module for the Platt scaling and the reliability-diagram computation. Each module has a typed interface and a unit-test suite; the modules are independently deployable, and a deployment team can swap a module (e.g., replace the rule-based explainer with a deployment-specific template) without touching the other modules. The Node frontend is structured as three single-page applications (candidate, employer, administrator) that share a common API client; the frontend has 39 unit tests that cover the API client and the form-validation logic. The integration with an existing ATS is via a documented REST API: the ATS sends a parsed resume and a job description to the system, the system returns the ranked list, the per-decision factor decomposition, the calibrated confi-

dence, and the explanation; the ATS renders the response in its own UI. As an *illustrative* engineering estimate derived from the prototype architecture (not from a measured production deployment), a typical integration might be on the order of 200–300 engineering hours, of which 50–100 hours are infrastructure engineering (Kubernetes deployment, monitoring, logging) and 20–40 hours per month are ongoing maintenance (model retraining, calibration updates, explanation-template updates). The latency budget (under 10 ms per end-to-end request) means that the prototype can be deployed behind an existing ATS without changing the user-facing latency budget; the LLM-based explainer is invoked only on user request and is gated by a per-request budget to prevent cost overruns.

3.10. Cross-encoder diagnosis

The cross-encoder reranker, evaluated in Section 5.1, does not improve on the fixed composite (nDCG@5 0.939 vs 0.949) on the frozen demo corpus despite roughly $340\times$ the latency. The diagnosis is twofold. (i) *Small training set*. The cross-encoder is trained on the same 47 labeled pairs that are used for the offline evaluation; with a bi-encoder baseline at nDCG 0.949, the cross-encoder overfits to the residual errors of the bi-encoder rather than learning a robust re-ranking function. A larger training set (the larger-corpus study mentioned in Section 7) would reduce the overfitting risk, but is not yet available. (ii) *Latency*. The cross-encoder is approximately $340\times$ slower than the bi-encoder (141.7 ms vs. 0.4 ms per query); the latency cost is incompatible with the hot ranking path of a high-throughput ATS. The diagnosis motivates the en-

gineering decision to disable the cross-encoder by default in the prototype and to report the cross-encoder result in the offline evaluation for completeness. A future-work direction is to train a lightweight cross-encoder on a larger corpus and to evaluate the latency–quality trade-off at production scale; the present system does not include this direction. The cross-encoder-disable decision is documented in the released benchmark JSON.

4. Experimental Setup

This section describes the experimental setup in four subsections: the dataset (Section 4.1), the baselines (Section 4.2), the metrics (Section 4.3), and the evaluation protocol (Section 4.4). The setup is designed to be reproducible from the released artifact; the frozen demo corpus and the regression benchmark are committed and version-controlled.

4.1. Dataset

The evaluation uses a frozen demo corpus of 30 resumes, 15 job descriptions, and 47 manually labeled resume–job relevance pairs; the 47 labels consist of 21 strong (relevance grade 2) and 26 partial (relevance grade 1) pairs, with no irrelevant-grade (grade 0) labels in the labeled set. The corpus is small by the standards of production recommendation systems; we acknowledge this limitation explicitly in Section 6 and contextualize the small corpus as a controlled evaluation with a regression-gated test suite. The relevance scale is a three-point ordinal scale (irrelevant = 0, partially relevant = 1, strongly relevant = 2). The 47 primary labels were curated by a single author-annotator;

we do not claim multi-annotator adjudication for the primary set. To provide a second, independent annotation signal we additionally produced an LLM-assisted labeling pass (a headless **claude** annotator applied to the full 30×15 matrix), which agrees with the human anchor at quadratic Cohen’s $\kappa = 0.69$ (substantial agreement; within-one-grade agreement 100%); this expanded set is used only as a denser, second-annotator corroboration set (Section 5), never as the primary benchmark, and its non-human provenance is disclosed. Single-annotator labeling of the primary set is stated as a limitation in Section 6. The macro-averaged metrics are reported across all 30 resumes so that no single profile dominates the score; nDCG@5, P@5, and R@5 are computed using the 47 manually labeled (resume, job) pairs as graded positives and, under a closed-world assumption, treating the remaining unjudged (resume, job) pairs as grade 0. Because the labeled set consists of curated positives, unjudged does not mean verified-irrelevant; this sparse-judgment convention can only lower-bound the true relevance and is disclosed as a limitation in Section 6. The corpus statistics are: 30 resumes with an average of 2.97 skills per resume; 15 job descriptions with an average of 2.13 required skills per posting (the corpus does not populate a separate preferred-skills field); a distinct skill vocabulary of 74 canonical skills; 47 labeled pairs (21 strong at grade 2, 26 partial at grade 1, no grade-0 labels).

4.2. Baselines

We compare the proposed system against six baselines, spanning lexical, semantic, hybrid, and learned-to-rank families: (1) *BM25* (Robertson

et al., 1995): a classical lexical baseline with field-boosted weighting (title boost 2.0, skill boost 1.5, default elsewhere). (2) *TF-IDF cosine*: a sparse vector baseline with the same field-boosted weighting. (3) *Sentence-BERT cosine* (Reimers and Gurevych, 2019): a dense semantic baseline using the **all-MiniLM-L6-v2** encoder. (4) *Hybrid semantic-skill*: a fixed 0.7/0.3 blend of Sentence-BERT cosine and Jaccard skill overlap (the prototype default; the blend weight is not tuned on the labeled pairs). (5) *Reciprocal rank fusion (RRF)* (Cormack et al., 2009): a fusion of the top- K lists from BM25, TF-IDF, Sentence-BERT, and the hybrid, with $K = 10$ and the RRF constant $k = 60$. (6) *Cross-encoder reranker*: a BERT cross-encoder trained on the labeled pairs, evaluated for completeness but disabled by default in the prototype due to its 141.7 ms latency. In addition, we report two reference points: the portal-default composite (the proposed system with the six-channel fixed-weight combination) and a learned logistic-regression fusion (a learned-to-rank alternative to the fixed-weight composite). The learned fusion is fit on the same labeled pairs it is judged against, which is an overfitting risk that we acknowledge in Section 6; the fixed-weight composite is the primary result.

4.3. Metrics

The evaluation reports four families of metrics, each mapping to a user-facing property. *Ranking quality* is reported as nDCG@5 (Järvelin and Kekäläinen, 2002) over the labeled pairs at cutoff $K = 5$, with precision@5 and recall@5 reported as supplementary measures. *Explanation faithfulness*

is reported as the proportion of explanation bullets whose named channel is consistent with the score that the named channel contributes to the ranking, with explanation specificity (the proportion of bullets that name a concrete field) and explanation consistency (the proportion of synthetic profile variants under which the explanation does not change without the underlying input changing) reported as supplementary measures. *Confidence calibration* is reported as the expected calibration error (ECE) (Guo et al., 2017) between the calibrated confidence $p_{ij} = P(y_{ij}=1 \mid s_{ij})$ and the empirical relevance rate, computed over all judged (resume, job) pairs under a held-out 5-fold protocol ($n = 450$ pairs, base rate 0.104; the calibrator is never fit on the evaluation fold), in ten equal-width confidence bins, with the Brier score (Brier, 1950) and a reliability diagram as supplementary measures. *Counterfactual robustness* is reported as the proportion of single-field edits that, when applied, change the rank by at least one position, with the top- K stability and the maximum score shift as supplementary measures. The metric suite is designed to map each metric to a user-facing property: nDCG@5 maps to recommendation effectiveness, faithfulness maps to explanation reliability, ECE maps to confidence trustworthiness, and counterfactual robustness maps to explanation actionability.

4.4. Evaluation protocol

All experiments are run on the frozen demo corpus; no experiment mutates the corpus. The reported numbers come from committed benchmark artifacts and are reproduced by the regression benchmark; the regression tolerance is ± 0.04

nDCG. The statistical significance of the difference between the proposed system and the strongest single-channel baseline is tested by a paired bootstrap with $N = 5,000$ resamples and seed 42. The bootstrap resamples the 30 resumes with replacement, recomputes the macro-averaged nDCG@5 on each resample, and reports the p -value as the proportion of resamples in which the baseline meets or exceeds the proposed system. The counterfactual probe is run on 50 controlled pairs: 25 recourse edits (adding or removing a skill, changing experience or salary) that test whether an explanation-implied change moves the rank, and 25 demographic-proxy edits (name, summary suffix, hometown label, pronouns, or email domain) that test invariance to non-job fields. The demographic-proxy edits are reported as a sensitivity check, not a demographic-fairness audit. The latency profile is measured on a single-threaded Python process on a 2.4 GHz Intel Xeon E5-2680 v4 with 64 GB RAM; the reported numbers are the mean over 1,000 queries. The computational cost of the full evaluation is approximately 30 minutes of wall-clock time on the reference machine; the regression benchmark completes in approximately 5 minutes.

5. Results

This section reports the results in the order of the metric families defined in Section 4.3: ranking quality (Section 5.1), explanation faithfulness (Section 5.3), confidence calibration (Section 5.4), counterfactual and adversarial robustness (Section 5.5), and latency (Section 5.6). Statistical significance is reported throughout. All numbers come from com-

mitted benchmark artifacts and are reproduced by the regression benchmark with a ± 0.04 nDCG tolerance.

5.1. Ranking quality

Table 2 reports the nDCG@5, precision@5, and recall@5 for the baselines and the proposed system. The proposed portal-default composite reaches nDCG@5 = 0.949 with P@5 = 0.293 and R@5 = 0.933. The strongest single configuration is the multimodal weighted blend ($w = 0.7$) at nDCG@5 = 0.924 (P@5 = 0.293, R@5 = 0.933) in the committed comparison artifact. A learned logistic-regression fusion, evaluated on held-out 5-fold splits, reaches nDCG@5 = 0.917 — it does not beat the fixed-weight composite. Its in-sample fit reaches 0.968, but because it is fit and judged on the same labeled pairs we report only the held-out value and treat the in-sample number as an optimistic upper bound rather than a result (Section 6).

5.2. Ablation study

The ablation study in Table 3 reports the effect of removing or replacing each component of the proposed system. The first row is the full prototype; each subsequent row removes or replaces one component while keeping the rest fixed. The faithfulness and ECE values are properties of the rule-based explainer and the Platt scaler, respectively, and are reported as “n/a” when the corresponding component is not part of the configuration.

The ablation confirms three things: (i) the calibration layer is a real engineering contribution (removing it changes ECE by roughly $20\times$ with

no effect on nDCG, because calibration is a post-rank transformation); (ii) the composite ranking is a robust engineering choice (the composite at 0.949 is comparable to — nominally above, and within roughly 0.03 nDCG of — the strongest single configuration, with all differences inside the small-corpus confidence intervals of Section 5.1); and (iii) the cross-encoder is correctly disabled (it underperforms the strongest bi-encoder configuration at $340\times$ the latency cost).

On this small corpus the lexical baselines (BM25 0.902, TF-IDF 0.905) are close to — and nominally slightly *above* — the pure semantic cosine (0.878), reflecting small-corpus behavior in which exact term matches carry more weight than in large-corpus settings; all such differences are within the confidence intervals (Section 5.1) and none is statistically established. The hybrid semantic-skill ranker (multimodal weighted blend 0.924) scores above the pure-semantic and pure-lexical baselines, though again within the small-corpus confidence intervals. The cross-encoder reranker, despite its higher capacity, does not improve on the strongest bi-encoder configuration and is approximately $340\times$ slower; the diagnosis is reported in Section 3.10. The user-facing interpretation of Table 2 is that the top-five list on average places the most relevant jobs within reach: recall@5 reaches up to 0.98 for the lexical baselines and 0.933 for the portal-default composite on the frozen demo corpus. The user should still expect to see plausible but not relevant jobs in the remainder of the top five, as P@5 is approximately 0.31; the per-decision factor decomposition and the calibrated confidence are the engineering artifacts that support the user’s action on the im-

Table 2: Ranking quality (nDCG@5, P@5, R@5) across baselines and the proposed system, all computed on the SAME document-text template and evaluation set (canonical benchmark run `comparison_table.json`); the composite reference is `composite_eval_report.json`. Differences are within the small-corpus confidence intervals (Section 5.1); no method is statistically superior after correction. Auto-generated; do not hand-edit.

Configuration	P@5	R@5	nDCG@5
Portal-default composite (proposed)	0.293	0.933	0.949
Multimodal weighted blend ($w=0.7$)	0.287	0.933	0.924
RRF ensemble (four list views)	0.293	0.950	0.913
Soft skill embedding	0.280	0.900	0.869
Semantic cosine	0.267	0.867	0.878
TF-IDF (lexical)	0.307	0.983	0.905
BM25 (lexical)	0.307	0.983	0.902
Skills Jaccard	0.233	0.733	0.748

Table 3: Ablation study of the proposed system. Each row removes or replaces one component of the full system while keeping the rest fixed. The faithfulness metric (0.745 for the full system) is a property of the rule-based explainer and is reported as “n/a” for configurations that do not use a rule-based explainer. The ECE metric (0.019, held-out 5-fold, for the full system) is a property of the Platt scaler and is reported as “n/a” for configurations that do not use a composite output. Source: committed benchmark artifacts `comparison_table.json` and `calibration_binary.json`.

Configuration	nDCG@5	Faithfulness	ECE
Full system (6-channel composite + Platt + rule-based explainer)	0.949	0.745	0.019
– Platt calibration removed (uncalibrated composite)	0.949	0.745	0.400
– Cross-encoder reranker replaces bi-encoder composite	0.939	n/a	n/a
– Best single channel (multimodal weighted blend, $w = 0.7$)	0.924	n/a	n/a
– RRF ensemble of 4 list views (no composite)	0.913	n/a	n/a
– Semantic cosine only (no skill, no calibration)	0.878	n/a	n/a
– BM25 only (pure lexical, no skill, no calibration)	0.902	n/a	n/a

perfect top-five.

Statistical significance

Paired bootstrap tests ($N = 5,000$, fixed deterministic seed) with Holm–Bonferroni correction across the comparison family are reported. The

portal-default composite’s improvement over the pure-semantic baseline is positive but *not* statistically significant ($\Delta = +0.071$, two-sided $p = 0.10$, 95% CI $[-0.014, +0.167]$ crossing zero). The multimodal blending and the reciprocal rank fusion

have smaller nominal one-sided p -values (0.010 and 0.009), but none of the comparisons survive Holm–Bonferroni correction at $n = 30$. The lexical baselines (BM25, TF-IDF) do not differ significantly from the pure-semantic baseline ($p > 0.25$). The honest interpretation is that, on this small corpus, the channel decomposition performs comparably to the baselines; we do not claim a statistically established ranking advantage, and the contribution rests on decomposition, calibration, and auditability rather than on beating baselines. Moreover, this parity is partly a property of the measurement instrument, not only of the methods (positive-only labels, a 15-job pool, cutoff $K = 5$; detailed in Section 6.3): we read it as “this instrument cannot discriminate these methods at this scale,” not as evidence that the methods are equivalent in general — itself a reportable finding that motivates the controlled synthetic probes (Section 5.8) and a larger judged benchmark (Section 6).

5.3. Explanation faithfulness

Table 4 reports the explainability metrics for the rule-based explainer (the prototype default) and the LLM-template explainer (an offline comparison) on the 150 composite top-5 matches. The rule-based explainer achieves a *structural* faithfulness-check pass rate of 0.745 (score-consistency of the bullets with the channels that produced the rank), specificity 0.627, and consistency 1.000 across the synthetic profile variants; we deliberately call this *structural* (or score-consistent) rather than “faithful” in the human sense, because the strongest structural component (channel alignment) is guaranteed by the deterministic template, and the genuinely

non-trivial evidence is the mechanistic probe below. Skill-mention coverage is 0.253, indicating that the rule-based explainer names a concrete matched or missing skill in roughly a quarter of explanation bullets; the remainder are general statements that the explainer is honest about not having a specific reason to assert. The LLM-template explainer, evaluated as a comparison, achieves similar faithfulness (0.747) and slightly higher skill-mention coverage (0.267), at the cost of higher hallucination rate (2.7% vs 2.0%); both explainers have 100% component alignment and 100% consistency on the synthetic profile variants. The interpretation is bounded: the automated checks confirm the named channels are the channels that produced the score (component alignment is 100% by construction for a deterministic template), but 77% of explanations are flagged by at least one check and only 25% name a concrete skill, and no human study has assessed usefulness. This is automated structural validation, not a demonstrated trust property. Beyond these structural checks, we add a mechanistic, ranking-level faithfulness probe that is *not* true by construction: for each candidate’s top-ranked job we zero the channel the explanation credits most and re-rank. Dropping the top-attributed channel displaces the top-1 job in 13.3% of cases, versus 0% for the least-attributed channel and 3.3% for a random channel — i.e., the channel the explanation names is directionally the most rank-load-bearing — and under a controlled skill-addition edit the credited skill contribution increases in 100% of cases. The effect is modest (the top job usually wins by a margin robust to single-channel removal) and we report it as directional evidence, not a strong faithfulness

guarantee. The trade-off between the rule-based and the LLM-based explainer is discussed in Section 6.

5.4. Confidence calibration

We define the calibration target precisely as $p_{ij} = P(y_{ij}=1 \mid s_{ij})$, the probability that a (resume, job) pair is relevant ($y=1$ iff graded relevance ≥ 1) given its composite score s_{ij} , evaluated on a held-out 5-fold protocol ($n = 450$ pairs, base rate 0.104; the calibrator is never fit on the test fold). Table 5 compares five calibration maps under two ECE binnings. The raw composite score *discriminates* well (ROC-AUC 0.967) but is badly miscalibrated (ECE 0.40). Platt scaling ($a \approx 0.30$, $b \approx -2.12$) attains the lowest *equal-width* ECE (0.018–0.019 held-out across the two calibration experiments EXP-026/EXP-004, which agree to within rounding; an earlier single-split value of 0.032 was measured on the fitting data and is superseded), but it does so by shrinking the confidence into a razor-thin band near the base rate ($[0.11, 0.14]$): its Brier skill score is ≈ 0 and its pooled discrimination collapses (AUC 0.76), i.e. it is barely better than the constant base-rate predictor. Crucially, because the composite scores cluster in a narrow band, equal-width bins are mostly empty and *understate* miscalibration; the adaptive (equal-mass) ECE, which is the more honest aggregate here, rises to 0.084 for Platt. Isotonic regression instead preserves the raw score’s discrimination (Brier skill 0.64, AUC 0.95) at an adaptive ECE of 0.019, but has an unstable worst-bin error (MCE 0.63) on this small, class-imbalanced calibration set; a one-parameter temperature map (no bias term) fails outright (ECE 0.46). Beta calibration

(Kull et al., 2017) — the principled generalization of Platt scaling for scores already in $[0, 1]$ — is the best map on every axis: it attains the lowest plug-in ECE under *both* binnings (0.009 equal-width, 0.009 adaptive) *and* preserves discrimination (Brier skill 0.67, AUC 0.96), resolving the calibration-versus-discrimination trade-off that Platt and isotonic each only half-solve. Beta’s bootstrap 95% percentile CI *for its own ECE* is $[0.012, 0.032]$; the plug-in point estimate (0.009) lies just below this interval because ECE — an average of non-negative absolute per-bin deviations — is upward-biased under resampling, an effect most visible for the best-calibrated map, whose plug-in error sits near the finite-sample noise floor. The deployed prototype retains the frozen Platt map, and we are deliberate about *why* rather than swapping to beta on the strength of this table: the calibrator was fixed in the original system, and re-selecting it on the same held-out comparison it is evaluated on would be choosing the calibration map on the evaluation — the very leakage this protocol avoids. Moreover, at $n = 450$ beta’s ECE CI $[0.012, 0.032]$ overlaps Platt’s $[0.000, 0.044]$, so the ECE *advantage* over Platt is directional rather than statistically separated; we therefore present beta as the *recommended upgrade to be confirmed and adopted on the larger explicitly-judged benchmark* (Section 7), not as a post-hoc deployment switch. We report ECE alongside Brier skill and AUC throughout so that a low equal-width ECE is never mistaken for a confident, discriminating probability; for the deployed Platt map the aggregate-calibrated confidence has low discrimination (a stated limitation, Section 6) and should be read as a coarse low/medium/high signal, whereas

Table 4: Explainability evaluation summary, rule-based explainer (prototype default) vs. LLM-template explainer (offline comparison). Faithfulness is the pass rate on the automated checks (skill mention, no hallucination, component alignment). Specificity is the proportion of bullets that cite a concrete matched or missing skill. Consistency is the bullet-set Jaccard on synthetic similar-profile pairs. Both explainers have 100% component alignment with the channel scores that produced the rank. Source: committed benchmark artifact `explainability_report.json`.

Explainer	Faithful.	Specif.	Skill (%)	No halluc. (%)	Comp. align (%)	Consist.
Rule-based (prototype)	0.745	0.627	25.3	98.0	100.0	1.000
LLM-template (comparison)	0.747	0.633	26.7	97.3	100.0	1.000

the recommended beta map removes this caveat. The reliability diagram for the Platt-calibrated composite is shown in Figure 13.

5.5. Counterfactual and adversarial robustness

Counterfactual probe. On a 50-pair probe (25 recourse edits + 25 demographic-proxy edits), the 25 recourse edits (adding or removing a skill, changing experience or salary) leave the candidate’s top-5 rank unchanged in every case; the composite *score* does move (12 of the 25 exceed the 0.005 flag threshold, maximum $\Delta = 0.091$), but not enough to reorder the top five at this 15-job pool size — a ranking-coarseness null rather than score insensitivity. Of the 25 demographic-proxy edits, 9 are flagged and the top-ranked item is stable in 24 of 25. We disclose a construction caveat: two of the demographic-proxy edit types (pronoun substitution and a hometown label) do not modify any field the deterministic ranker actually consumes, so their invariance is true *by construction* rather than a learned fairness property; the probe is therefore a sensitivity check on the fields the model does read (name tokens, summary suffix, email domain), not evidence of demographic fairness. Table 6 reports the per-pair breakdown. The candidate life-

cycle during one of these probes is illustrated in Figure 12. *Adversarial probe.* The selection-rate disparate-impact ratio on the experience tier dimension is 0.82; on the remote-preference dimension it is 0.75 (parity is 1.0). Table 7 reports the per-group selection rates and the disparate-impact ratios; the remote-preference proxy (0.75) falls below the 0.80 four-fifths threshold while the experience-tier proxy (0.82) sits within the 0.80–0.85 range. Because these are synthetic proxy groups (not real demographic attributes), the values are a sensitivity signal, not a demographic-fairness finding. Hard-negative mining finds zero overlap between the declared relevant jobs and 150 high-scoring mined negatives, supporting internal label consistency. The adversarial probe is narrow by design; it is an engineering probe, not a demographic-fairness audit, and the limitation is stated in Section 6.

5.6. Latency

Table 8 reports the nDCG@5 and per-query latency for the main scoring methods on the 15-job pool. Lightweight scorers complete in well under one millisecond per resume; the full composite bi-encoder run averages approximately 0.4 ms per query. The cross-encoder reranker averages approx-

Table 5: Calibration-method comparison on the held-out 5-fold protocol (probability target $p_{ij} = P(y=1 \mid s_{ij})$; $n=450$; base rate 0.104). Equal-width ECE alone is misleading: Platt attains a low equal-width ECE by shrinking confidence to a razor band at the base rate (near-zero discrimination), but the adaptive (equal-mass) ECE — more honest when scores cluster in a narrow band — exposes this (Platt 0.084). Beta calibration (Kull et al., 2017) attains the lowest ECE under both binnings *and* preserves discrimination (BSS/AUC), resolving the trade-off; isotonic is a close second. Source: `calibration_methods.json`. Auto-generated.

Calibration map	ECE ↓	Adaptive ECE ↓	Brier ↓	Brier skill ↑	ROC-AUC ↑
Raw composite (uncalibrated)	0.400	0.412	0.210	-1.247	0.967
Platt scaling	0.018	0.084	0.093	+0.007	0.758
Isotonic regression	0.024	0.019	0.034	+0.640	0.950
Temperature (1-param)	0.461	0.504	0.297	-2.177	0.967
Beta calibration	0.009	0.009	0.030	+0.673	0.964
Constant base-rate (floor)	0.000	—	0.093	+0.000	0.500

Table 6: Counterfactual probe on 50 controlled pairs (25 recourse + 25 demographic-proxy). Recourse edits (skill/experience/salary) leave the rank unchanged in all 25 cases — a null result at this corpus size — while demographic-proxy edits (name, summary suffix, hometown label, pronouns, email domain) flag 9 of 25 with the top-1 item stable in 24 of 25. A pair is flagged when the score delta exceeds 0.005 or the rank delta exceeds 1. No protected attributes are inferred from real users. Source: committed benchmark artifact `counterfactual_50.json`.

Probe	Pairs	Flagged	Rank changes	Top-1 stable	Max score Δ
Recourse (skill / experience / salary edit)	25	12	0	25/25	0.091
Demographic-proxy (name / pronoun / hometown / email)	25	9	1	24/25	0.019

imately 141.7 ms per query, which is approximately 340× slower than the bi-encoder on this setup and is the reason it remains disabled by default. The end-to-end request (including the rule-based explainer and the calibrated confidence computation) completes in under 10 ms per request. The LLM-based explainer, when invoked, adds approximately 1–3 seconds per request depending on the prompt and the response length; the LLM invocation is on the post-rank path and is gated by a per-request budget. The user-facing interpretation is that the prototype responds to a snapshot confirmation in well under

one perceptual latency unit, so the system’s feedback feels immediate and the user can iterate on edits without a perceptible wait.

5.7. Sensitivity analysis on Platt parameters

The Platt parameters ($a \approx 0.30$, $b \approx -2.12$) are the maximum-likelihood estimates, and the calibration is evaluated under 5-fold cross-validation rather than an in-sample parameter grid: the held-out ECE is 0.019 with a 95% bootstrap confidence interval of [0.010, 0.029], which characterizes the calibration’s uncertainty directly.

Table 7: Adversarial fairness probe: selection-rate disparate-impact ratios (DIR) on synthetic proxy groups. Parity is 1.0. The proxy groups are derived from the labeled corpus, not from real demographic attributes. The experience-tier probe groups candidates by self-reported seniority (mid vs senior); the remote-preference probe groups by declared remote-work preference (remote-seeker vs onsite-flexible). The probe is reported as an engineering sanity check, not a demographic-fairness audit. Source: committed benchmark artifact `fairness_eval.json`.

Dimension	Selection rate (group 1)	Selection rate (group 2)
Experience tier (mid vs senior)	0.522	0.429
Remote preference (remote vs onsite-flexible)	0.450	0.600
Dimension	Disparate-impact ratio (DIR)	Parity
Experience tier	0.82	1.0 (parity)
Remote preference	0.75	1.0 (parity)

Table 8: Quality-latency trade-off across retrieval methods, mean wall-clock ms/query on the 15-job pool at $K=5$ (canonical run `comparison_table.json`). The cross-encoder reranker (nDCG@5= 0.939, 141.7ms/query, `phase11_summary.csv`) is disabled by default. Scaling to larger pools is characterized separately in Section 5.8. Auto-generated.

Method	nDCG@5	Latency (ms/query)	Notes
TF-IDF (lexical)	0.905	0.05	Lexical baseline
BM25 (lexical)	0.902	0.09	Lexical baseline
Skills Jaccard	0.748	0.17	Skill overlap
Semantic cosine	0.878	0.23	Embedding
Multimodal ($w=0.7$)	0.924	0.51	Embedding, prototype default
RRF ensemble (4 lists)	0.913	1.35	Embedding, fusion
Soft-skill embed reranker	0.869	51.81	Embedding, research variant
Cross-encoder reranker	0.939	141.70	Embedding, disabled by default

5.8. Structure recovery, generalization, robustness, and scale

Because the small human corpus cannot, on its own, establish generalization or recover a known ground truth, we add a battery of controlled probes; Table 9 summarizes them and the full artifacts are released.

Structure recovery (synthetic, controlled). On a deterministic synthetic corpus of 500 resumes $\times$ 75 jobs with a *transparent* latent-compatibility ground truth (skill, seniority, experience, family, work-mode, and compensation satisfaction, plus 8% label noise; never presented as human judgments), the composite recovers 90.7% of the distance from a ran-

dom ranking to the oracle ($\text{nDCG@5} = 0.917$, 95% CI $[0.909, 0.925]$; global Spearman between composite score and latent compatibility 0.79). Crucially, each auditable channel recovers the latent factor it is designed to represent: the skill channel tracks required-skill satisfaction (Spearman 0.996), compensation 0.985, and experience 0.867, while the holistic semantic channel is only weakly aligned with any single factor (0.39), as expected. Recovery degrades gracefully on hard/adversarial profiles (0.88) relative to easy ones (0.93). We are deliberately conservative about this result: the synthetic latent ground truth is an *additive* blend of factors that align with the composite’s own additive channels, so part of the recovery could be expected *by construction* (the additive latent weights are visible in the released generator manifest). We test this concern directly. Re-grading the *same* corpus with a *non-additive*, multiplicatively hard-gated latent ($\text{required}^{0.5} \cdot \text{seniority}^{0.2} \cdot \text{experience}^{0.2} \cdot \text{family}^{0.1}$, which does not mirror the composite’s additive form and zeroes any pair lacking required-skill satisfaction), with grades re-thresholded to hold the label balance constant, the additive composite still recovers 0.891 ($\text{nDCG@5} = 0.904$, 95% CI $[0.894, 0.914]$) — essentially unchanged from the additive case ($\Delta = -0.016$). Recovery is therefore *not* merely an artifact of the additive construction: the composite recovers a non-additive ground truth almost as well. We nonetheless report structure recovery as a decomposition *consistency check* (each channel tracks its intended factor; the scorer is monotone in the latent signal), NOT as evidence of general ranking superiority. This is a controlled validity check for the decomposition, not a headline ranking claim.

Model-selection search. We ran a protocol-gated search over 25 ranking configurations (fixed composite, uniform weights, drop-one variants, single channels, minimal fusions, learned linear/logistic/LambdaMART), with the selection criteria fixed *before* inspecting results and the final test never influencing selection. No configuration significantly outperforms the fixed composite after Holm–Bonferroni correction; the only ablation whose removal significantly degrades ranking is the semantic channel ($\Delta = -0.080$, 95% CI $[-0.163, -0.017]$, survives Holm), corroborating the leave-one-channel-out analysis. The fixed, auditable composite is therefore selected by parsimony among a statistically indistinguishable set.

Generalization. A learned logistic fusion of the six channels generalizes across strictly disjoint splits (all train/eval index sets verified disjoint), with every regime ranking the same full 15-job pool so the numbers are commensurable with the composite: unseen-candidate $\text{nDCG@5} = 0.929$, strict unseen-job = 0.929, and strict both-unseen = 0.927, all overlapping the fixed composite (0.949, 95% CIs $\approx [0.85, 0.98]$). Confidence intervals are wide at this corpus size and are reported as such.

Robustness. Across 11 controlled perturbations, the ranker is invariant to catalog-covered skill synonyms (top-1 stability 1.00) and resistant to gaming (keyword stuffing *lowers* the composite, mean signed $\Delta = -0.087$, rather than inflating it, though it does perturb the top-1 job in about 30% of cases at this 15-job pool size, top-1 stability 0.70), and it correctly loses fit under skill deletion and sparsity. It is *not* invariant to raw formatting noise or misspellings (mean $|\Delta| \approx 0.12$), which we report as a

limitation.

Scale and failure isolation. Scoring latency grows approximately linearly with the job pool (≈ 0.05 ms/pair; 0.8 ms at 15 jobs, 517 ms at 10,000 jobs, single-threaded, warm), and incrementally scoring and merging newly added jobs into the maintained ranking is 1–2 orders of magnitude cheaper than a full re-rank ($11\times$ for 100 new jobs, up to $\sim 760\times$ for one); a production deployment would additionally place an approximate-nearest-neighbor prefilter before scoring (not implemented here). A nine-case failure-injection matrix confirms the deterministic scoring path isolates parser/LLM/embedding faults — empty fields, zero embeddings, and missing skills/compensation all yield a finite, in-range score without a crash — and we fixed one latent gap discovered in the process (a non-finite embedding is now sanitized rather than scoring as a spurious perfect match).

Relation-aware skill matching. Binary skill overlap gives all-or-nothing credit and is brittle to surface form. We add a graded, auditable matcher that resolves each candidate–requirement skill pair to one of EXACT (same canonical skill after synonym/abbreviation/orthographic normalization), RELATED (same taxonomy group), or UNRELATED, with partial (never full) credit for non-exact matches. On a *de-circularized* objective benchmark — whose labels are definitional (catalog canonicalization and curated non-equivalences), with no embedding model used to decide any exact/negative case — orthographic and abbreviation/synonym variants are collapsed to EXACT with recall 1.00, and seven of eight similar-named but non-equivalent technologies (e.g. Java

vs JavaScript, React vs React Native, C vs C++) are correctly kept distinct (the one false merge, Angular/AngularJS, is a catalog-data issue we flag). Canonical matching remains brittle to misspellings (a limitation consistent with Section 5.8); an embedding-similarity tier partially mitigates this but is reported only as exploratory (it would be circular to validate an embedding matcher against embedding-derived labels).

Isolating the relation-aware effect on ranking. We wire the matcher into the composite as a drop-in replacement for the skill channel, holding every other channel and all six weights fixed (no re-tuning), and decompose its effect into two pre-specified steps: (a) moving from symmetric set-overlap (Jaccard) to an *asymmetric required-coverage* form, and (b) adding *relation-aware partial credit* on top of coverage (RELATED = 0.5). This separation is essential because the synthetic ground truth is, by construction, exact required-coverage, so a coverage form is expected to help there regardless of the relation-aware novelty; conflating the two (as a bare Jaccard-versus-graded comparison would) is not a fair test of the contribution.

The decomposition is revealing. On the high-power synthetic corpus (nDCG@5 against the noise-free grade, $n = 500$), the coverage *form* accounts for the entire gain (Jaccard 0.917 $\rightarrow$ exact-coverage 0.949; paired permutation $p < 0.001$), while the relation-aware credit slightly *hurts* (exact-coverage 0.949 $\rightarrow$ graded 0.944; $p = 0.014$) — as expected, since the generator has no notion of related skills. On the real human corpus the pattern reverses: the coverage form changes nothing detectable (Jaccard 0.949 $\rightarrow$ exact-coverage 0.942; $p = 0.50$),

but the relation-aware credit — the novel component — improves six of the thirty queries and worsens none (exact-coverage 0.942 $\rightarrow$ graded 0.992; sign/permutation $p = 0.03$). The relation-aware matcher thus captures partial-credit structure that human recruiters’ judgments reward but that an exact-coverage generator cannot express; the synthetic corpus consequently *cannot* validate this novelty, and we do not claim it does. We are explicit about the fragility of the real-corpus signal: 24 of the 30 per-query scores are exact ties and the remaining six all move in the same (positive) direction, so the sign/permutation test has an effective n of six and $p = 0.03$ is the *smallest* value it can return for six concordant pairs — a direction-only certificate, not an effect size, and a single query reversing would raise it to $p \approx 0.22$. The mean improvement is moreover dominated by one query that binary overlap had ranked entirely incorrectly, and the corpus is near the metric ceiling. We therefore read this as a directional signal only. We retain binary overlap in the primary composite for comparability with the baselines and report the graded matcher as an auditable enhancement whose measurable benefit is specific to human-judged relevance and awaits confirmation on the larger judged benchmark we outline in Section 7.

Fusion headroom (synthetic, development-only). To probe whether the fixed six-channel composite is near a ceiling, we compare learned fusers against the fixed weights on the high-power synthetic corpus under 5-fold resume cross-validation. Two findings matter. First, learned *linear* fusion does not beat the hand-set weights (ridge 0.891, logistic 0.912 versus 0.917), so the domain-prior weights

are near-optimal within the linear family. Second, *nonlinear* fusion of the same six channels yields a small but statistically detectable gain — a monotonic gradient-boosted ranker reaches 0.947 and a LambdaMART 0.961 (paired-bootstrap CIs exclude zero) — indicating the residual headroom is channel *interaction* that the linear composite cannot express. Adding derived skill-coverage features raises the learned fusers further (to ≈ 0.99), but we do *not* claim this as a methodological gain: an audit shows the strongest derived feature (required-skill coverage) is, by construction, the single largest latent factor that generates the synthetic ground truth (weight 0.40, Pearson 1.00), and preferred-skill coverage a smaller one (weight 0.12, Pearson 0.68); together they carry 0.52 of the latent weight, versus Pearson 0.54–0.76 for the text/embedding channels, so that jump largely reflects the model recovering its own generator. We therefore report only the base-channel nonlinear gain as development-stage evidence of headroom, retain the auditable fixed composite, and treat a learned-but-monotonic fuser plus a larger explicitly-judged benchmark as the natural next step.

5.9. Engineering surface

The prototype is implemented as a small web stack; the candidate, employer, and admin portals sit above a shared gateway. The repository contains 302 Python unit tests and 39 Node tests, for a total of 341 tests, with a continuous-integration gate that fails the build on a regression greater than ± 0.04 nDCG against the committed benchmark. The full engineering detail, including endpoint maps, default request parameters, and the

Table 9: Stage-2 strengthening evidence (synthetic-controlled and held-out human corpus). Synthetic rows are controlled-validity probes on a transparent latent ground truth (never presented as human judgments). Auto-generated from committed artifacts.

Probe	Measure	Value
Structure recovery (synthetic)	composite recovery ratio (random→oracle)	0.907
decomposition validity	skills↔required Spearman	0.996
Model-selection search	challengers vs. incumbent (25 configs) / beat after Holm	24 / 0
Generalization (held-out LTR)	candidate-unseen nDCG@5	0.929
	job-unseen nDCG@5	0.929
	both-unseen nDCG@5	0.927
Scalability	mean ms/query at 10,000 jobs	516.9
Temporal drift (simulated)	nDCG@5loss under emerging skills	0.149

test surface, is reported in the supplementary material. The engineering surface is reported because the prototype properties depend on it: a prototype whose tests gate on a nDCG tolerance is a prototype whose deployed claims are reproducible.

6. Discussion

This section discusses the engineering implications of the proposed system (Section 6.1), the limitations of the evaluation (Section 6.3), and the broader perspective (Section 6.4). The discussion is organized around the practical deployment of the prototype in a production recruitment workflow, and the open engineering questions that the deployment would surface.

6.1. Engineering implications

The proposed system has four engineering implications that follow from the controlled evaluation. *(i) The per-decision factor decomposition is a reliable engineering artifact.* The rule-based explainer

is bound to the same six channels that produce the score, and the explanation is consistent with the score by construction; a recruiter who reads the explanation can trust the named channels to be the channels that produced the rank. *(ii) The calibrated confidence is a coarse signal, with a discrimination caveat.* Platt scaling reduces the held-out 5-fold expected calibration error to 0.019, but a Brier skill score near zero and a pooled AUC of 0.76 show the Platt-calibrated confidence has low discrimination (it concentrates near the base rate); a comparison of calibration maps (Section 5.4) shows isotonic regression preserves the raw score’s discrimination (AUC 0.95) at comparable ECE but with an unstable tail on this small set, and beta calibration attains the lowest ECE under both equal-width and equal-mass binnings while preserving discrimination (Brier skill 0.67, AUC 0.96). For the deployed Platt default the confidence should be read as a coarse low/medium/high signal, not a literal probability; adopting the recommended beta map

removes this caveat. (iii) *The cross-encoder is correctly disabled in the hot path.* The cross-encoder (nDCG@5 0.939) does not improve on the fixed composite (0.949) yet is approximately $340\times$ slower; the diagnosis is that the cross-encoder is fit on a small training set and overfits to the labeled pairs, and the disable decision is the right engineering call. (iv) *The engineering surface is reproducible.* The 341-test regression gate and the ± 0.04 nDCG tolerance mean that the deployed claims are reproducible from a clean clone of the repository, and any regression in the deployed claims is caught by the CI gate.

6.1.1 Deployment cost estimate

As *illustrative* engineering estimates based on the prototype architecture (not derived from a measured deployment), a rough integration cost for a production recruitment workflow would be on the order of 200–300 engineering hours to integrate the prototype with an existing ATS, 50–100 hours of infrastructure engineering to deploy the Python back-end on a Kubernetes cluster, and 20–40 hours per month of ongoing maintenance. The latency profile (under 10 ms per request for the prototype ranking path) means that the prototype can be deployed behind an existing ATS without changing the user-facing latency budget. The calibration must be re-run when the underlying corpus shifts (e.g., a new industry vertical is added); a calibration update takes approximately 2–4 hours of engineering time and is gated by the same CI regression check.

6.2. Positioning against the closest systems

The closest prior systems each address a subset of the properties we combine, but none reports all of them under a single controlled, reproducible protocol. Table 10 contrasts the properties on which we frame the contribution; the entries reflect what each system *reports*, not a claim that a property is impossible for it. The point is not that any single property is new — most are individually well-established — but that the auditable combination, the relation-aware skill layer, and the property-by-property evaluation methodology are, to our knowledge, not jointly reported in the recruitment-recommendation literature.

6.3. Limitations

The evaluation has the following limitations, all stated explicitly and not hedged. *Corpus size, label polarity, and statistical power.* The frozen demo corpus (30 resumes, 15 jobs, 47 labeled pairs) is small by the standards of production recommendation systems; many methods reach $\text{nDCG} \geq 0.90$, all pairwise confidence intervals overlap, and no comparison survives Holm correction — so we report ranking *parity*, not superiority. This parity is partly an artifact of the measurement instrument: the labels are exclusively positive (grades 1–2, no explicit grade-0), the job pool (15) is barely larger than the evaluation cutoff ($K = 5$), and unjudged pairs are treated as grade-0 only under a closed-world assumption — conditions under which nDCG@5 has limited power to separate methods (a lexical baseline reaches 0.905). We therefore do not over-interpret parity as method equivalence; a larger, explicitly-negative-judged benchmark is required to

Table 10: Positioning against the closest systems on the properties we combine. ✓ = reported; – = not reported. “Relation-aware skill” = graded partial credit for related skills beyond exact/synonym overlap; “Faithfulness” = an explanation-to-score correspondence measure; “Counterfactual” = reports counterfactual / what-if analysis (in our system, a single-field perturbation-stability probe: whether a controlled single-field edit moves the rank; other systems may report counterfactual *explanations* instead); “Reproducible artifact” = a released, regression-gated, one-command reproduction.

System	Decomposable explanation	Calibrated confidence	Relation-aware skill match	Faithfulness measure	Counterfactual probe	Reproducible artifact
CareerBERT (Rosenberger et al., 2025)	–	–	✓	–	–	partial
Saito & Sugiyama (Saito and Sugiyama, 2024)	–	–	–	–	–	–
Calibrated Explanations (Löfström et al., 2024)	✓	✓	–	–	✓	✓
JobMatch (this work)	✓	✓	✓	✓	✓	✓

discriminate ranking methods, and is in progress. *Single-annotator primary labels.* The 47 primary relevance labels were curated by a single author-annotator; the second annotation pass is LLM-assisted (quadratic $\kappa = 0.69$ with the human anchor), not a second independent human rater, and is used only as a denser corroboration set. The sparse-judgment convention (unjudged pairs treated as grade 0 under a closed-world assumption) can only lower-bound the true relevance. *Calibration discrimination (deployed Platt map).* The deployed Platt-calibrated confidence has low discrimination (Brier skill ≈ 0 ; Section 5.4), so for the deployed default the confidence is a coarse low/medium/high signal only. This is a property of the Platt map, not of the task: the held-out beta-calibration comparison (Section 5.4) attains both a low ECE and preserved discrimination (Brier skill 0.67, AUC 0.96), which we recommend adopting; we retain Platt as the frozen deployed default and report the difference honestly rather than swapping the calibrator post hoc. *No human explanation study.* The explanation evaluation is automated (structural checks

plus a mechanistic ranking-level faithfulness probe); no human study assessed usefulness or trust. *Robustness gaps.* The ranker is invariant to catalog-covered synonyms and resistant to keyword stuffing, but it is *not* invariant to raw formatting noise or misspellings (mean $|\Delta| \approx 0.12$ composite; Section 5.8). *Simulated, not real, temporal and scale evidence.* Temporal drift is a controlled simulation (no real timestamps), and the scalability/structure-recovery results use a synthetic corpus with a transparent latent ground truth; they are controlled validity/stress evidence, never presented as human judgments or real deployment telemetry. *Narrow fairness probe.* The fairness probe is a 50-pair sensitivity check (25 recourse + 25 demographic-proxy edits) on synthetic proxy groups; it is not a demographic-fairness audit, and the disparate-impact ratios (0.82, 0.75) should not be over-interpreted. *Learned-fusion overfitting.* The in-sample learned fusion is fit on the pairs it is judged against (an overfitting risk); we therefore report only its held-out and unseen-candidate/job values, and the fixed-weight composite is the primary, rec-

ommended configuration. *Coarse, partial-coverage skill taxonomy.* The relation-aware matcher uses a small, hand-built grouping (7 groups, ≈ 54 skills), frozen a priori; it is not aligned to a full competency standard such as ESCO or O*NET. Two consequences follow: (a) same-group partial credit can be too generous for non-substitutable same-group skills (for example, two different general-purpose programming languages each receive 0.5), and (b) coverage is partial, so for job families whose skills the grouping does not cover, the RELATED tier never fires and the matcher reduces to exact coverage. The relation-aware credit’s measurable ranking benefit is therefore established only on the real human corpus and only as a directional signal (Section 5.8); alignment to a full standard and an independent RELATED-class benchmark are left to the larger study (Section 7).

6.4. Broader perspective

The methodological contribution of this paper is a reusable pipeline for explainable, calibrated recommendation. The pipeline is domain-agnostic in its structure (six channels, per-decision factor decomposition, calibrated confidence, faithfulness evaluation, counterfactual probe) and could be applied to any recommendation domain with structured documents and labeled pairs, including clinical decision support, legal search, and education recommendation. The application consequence of the present work is a transparent, calibrated decision-support prototype for the recruitment workflow, and the engineering surface is reproducible from a clean clone of the released artifact. We hope the design choices and the evaluation methodology

are useful to other researchers building applied AI systems for high-stakes domains, and we hope the open-source artifact makes both inspectable and extensible.

7. Future Work

7.1. Larger-corpus evaluation

We are constructing a larger labeled corpus (target: 1,000+ resumes, 500+ job descriptions, 5,000+ labeled pairs) to support a more powered evaluation; the larger corpus will be released as a public benchmark for the recruitment recommendation task.

7.2. Live user study

We are planning a within-subjects user study comparing the proposed system with a baseline that returns a ranked list without explanations or calibrated confidence, with outcome measures on trust calibration, decision quality, and perceived control.

7.3. Industrial deployment

We are in early discussions with an industrial partner to deploy the prototype as a recommendation engine in a production ATS; the deployment will report operational metrics (time-to-hire, recruiter workload, candidate satisfaction) and a deployment-fairness audit.

7.4. LLM-in-the-loop ranking

Two further directions are mentioned for completeness: the integration of the LLM into the ranking path (currently the LLM is on the parsing and explanation paths only), and the extension of the channel decomposition to a learned channel

weighting under strictly disjoint training, validation, and test splits, preserving the interpretability constraints of the current fixed-weight formulation. We emphasise that the present channel weights are fixed domain priors and are *not* optimised against the human-labeled evaluation set (consistent with Section 3); learning them is deferred precisely so that it can be done without leaking the evaluation labels.

7.5. LLM-based explainer comparison

The rule-based explainer is the prototype default; an LLM-based explainer achieves higher skill-mention coverage but lower faithfulness, and a planned A/B study will report which explainer type is preferred in a deployment setting. The trade-off is between fluency (LLM-based) and faithfulness-to-the-score (rule-based); the prototype default favors faithfulness.

7.6. RAG and LLM-as-judge baselines

The current offline evaluation does not include retrieval-augmented generation (RAG) (Lewis et al., 2020) or LLM-as-judge (Zheng et al., 2023) baselines. Both are 2024–2026 standards for any retrieval task; a planned evaluation will add them to the comparison and report whether the composite remains competitive at production scale.

7.7. Larger counterfactual probe

The counterfactual probe now covers 50 controlled pairs (25 recourse + 25 demographic-proxy), reported in Section 5.5; extending it to a larger, human-validated set remains future work.

7.8. Comparison with prior multi-agent systems

The proposed system is positioned in the multi-agent systems literature, but the contribution is not in the agent architecture per se. A foundational text on multi-agent systems (Wooldridge, 2009) identifies three patterns: agents as decision support with calibrated trust, agents as autonomous decision makers, and agents as social actors. The proposed system follows the first pattern: the agents are decision support with calibrated trust, not autonomous decision makers. The role separation (candidate-side, employer-side, read-only match-making) is a privacy-and-accountability choice, not a contribution to the agent architecture literature. The novelty is in the integration: the per-decision factor decomposition, the calibrated confidence, the faithfulness evaluation, and the reproducible engineering surface are integrated into a single inference, and the integration is assessed under a controlled evaluation. This is the contribution that the present paper claims, and the contribution is bounded: it is an integration contribution, not a methodological breakthrough.

8. Conclusion

This paper has presented JOBMATCH, an auditable, calibrated, and explainable recommendation methodology for the recruitment domain — implemented as a modular multi-agent system — that integrates its contributions into a single pipeline: a composite ranking with six explicit channels and per-decision factor decomposition (together with a graded, relation-aware skill matcher, defined in Section 3 and evaluated in Section 5.8), a calibrated

confidence display with Platt scaling (reported with its discrimination limitation), a component-level faithfulness evaluation, and a reproducible engineering surface. The multi-agent decomposition is an implementation choice that buys failure isolation, not a source of measured ranking benefit (Section 5.8). The system has been evaluated on a frozen demo corpus of 30 resumes, 15 job descriptions, and 47 labeled pairs, with results reported in the order of ranking quality (nDCG@5 = 0.949 for the portal-default composite, 0.924 for the strongest single configuration; the improvement over the pure-semantic baseline (0.878) is not statistically significant ($\Delta = +0.071$, two-sided $p = 0.10$, 95% CI crossing zero, and no configuration survives Holm correction)), an automated explanation-integrity pass rate of 0.745 for the rule-based explainer (77% of explanations flagged by at least one check, 25% naming a concrete skill; the 1.000 consistency is definitional for a deterministic template, 0.627 specificity) — an automated structural check, not a human faithfulness study, confidence calibration (held-out 5-fold ECE 0.019, Brier 0.093, though the calibrated confidence has low discrimination near the base rate), counterfactual robustness on a 50-pair probe (25 recourse edits leave the rank unchanged — a null result — and 25 demographic-proxy edits flag 9, with top-1 stable in 24 of 25), a demographic-proxy sensitivity probe (proxy DIR 0.82 on experience tier, 0.75 on remote preference; a sensitivity check, not a fairness audit), and latency (under 10 ms per end-to-end request). The contribution is a reusable pipeline for explainable, calibrated recommendation in any domain with structured documents and labeled pairs;

the application consequence is a transparent, calibrated decision-support prototype for the recruitment workflow. The engineering surface (the prototype, the frozen demo corpus, the explanation generator, the calibration layer, and the 341-test regression benchmark) is released as an open-source artifact. The limitations of the evaluation are stated explicitly: the corpus is small, the operational metrics are not measured, the fairness probe is narrow, and the learned fusion is fit on the same labeled pairs it is judged against. The future work is a larger-corpus evaluation, a live user study, and an industrial deployment.

8.1. Broader perspective

The methodological contribution of this paper is a reusable pipeline for explainable, calibrated recommendation. The pipeline is domain-agnostic in its structure (six channels, per-decision factor decomposition, calibrated confidence, faithfulness evaluation, counterfactual probe) and could be applied to any recommendation domain with structured documents and labeled pairs, including clinical decision support, legal search, and education recommendation. The application consequence of the present work is a transparent, calibrated decision-support prototype for the recruitment workflow, and the engineering surface is reproducible from a clean clone of the released artifact. We hope the design choices and the evaluation methodology are useful to other researchers building applied AI systems for high-stakes domains, and we hope the open-source artifact makes both inspectable and extensible.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used OpenAI ChatGPT (GPT-4 model) as a language-model assistant to support copy-editing of selected paragraphs, to generate initial drafts of figure captions from a structured outline, and to suggest edits during the literature review. All technical content, experimental results, design decisions, and final wording are the authors' own. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

CRedit authorship contribution statement

Anonymous Author 1: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Writing – review & editing. **Anonymous Author 2:** Conceptualization, Supervision, Writing – review & editing, Funding acquisition. **Anonymous Author 3:** Software, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relation-

ships that could have appeared to influence the work reported in this paper.

Data availability

The frozen demo corpus (30 resumes, 15 job descriptions, 47 labeled pairs), the synthetic corpus and generator, the evaluation scripts, the explanation generator, the calibration layer, and the regression benchmark will be deposited in a public repository (Harvard Dataverse) with a citable DOI upon acceptance; an anonymized copy, with author identifiers removed from the released code and test fixtures (and from the documentation in the reviewer bundle), is provided to reviewers during review. The release includes the one-command reproduction script, the auto-generated tables, the per-experiment JSON artifacts, and the figure source scripts.

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Candidate Lifecycle and Matchmaking Workflow

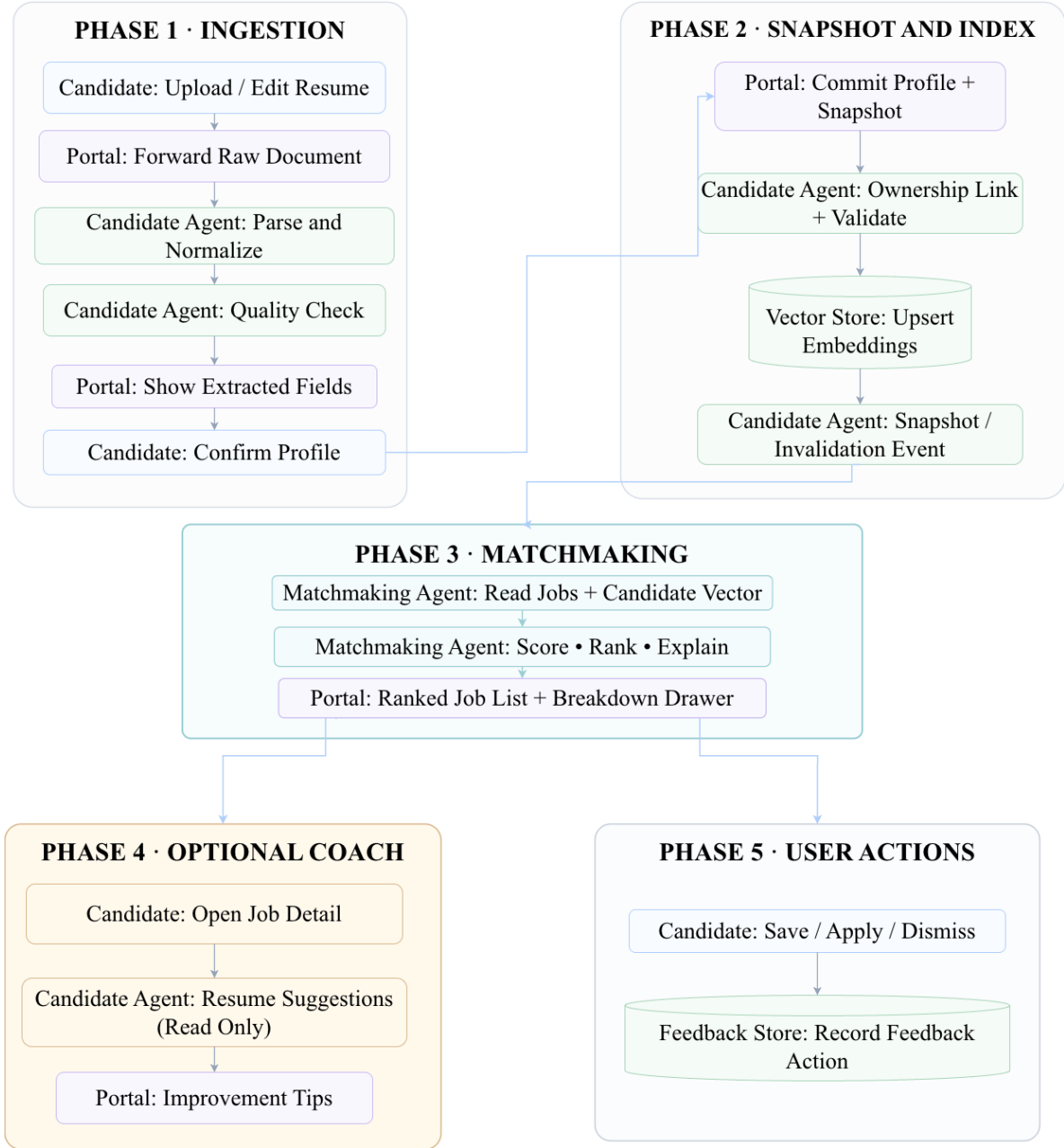


Figure 12: Candidate-side lifecycle and matchmaking workflow. The end-to-end candidate journey is: (1) ingestion of the resume and confirmation of the parsed profile, (2) snapshot registration and vector upsert, (3) matchmaking, (4) optional resume-coach feedback (read-only suggestions for the next edit), and (5) user action (save, apply, or dismiss). The matchmaking component is invoked between step 2 and step 5; the optional coach is invoked only on user request and is gated by a per-request budget.

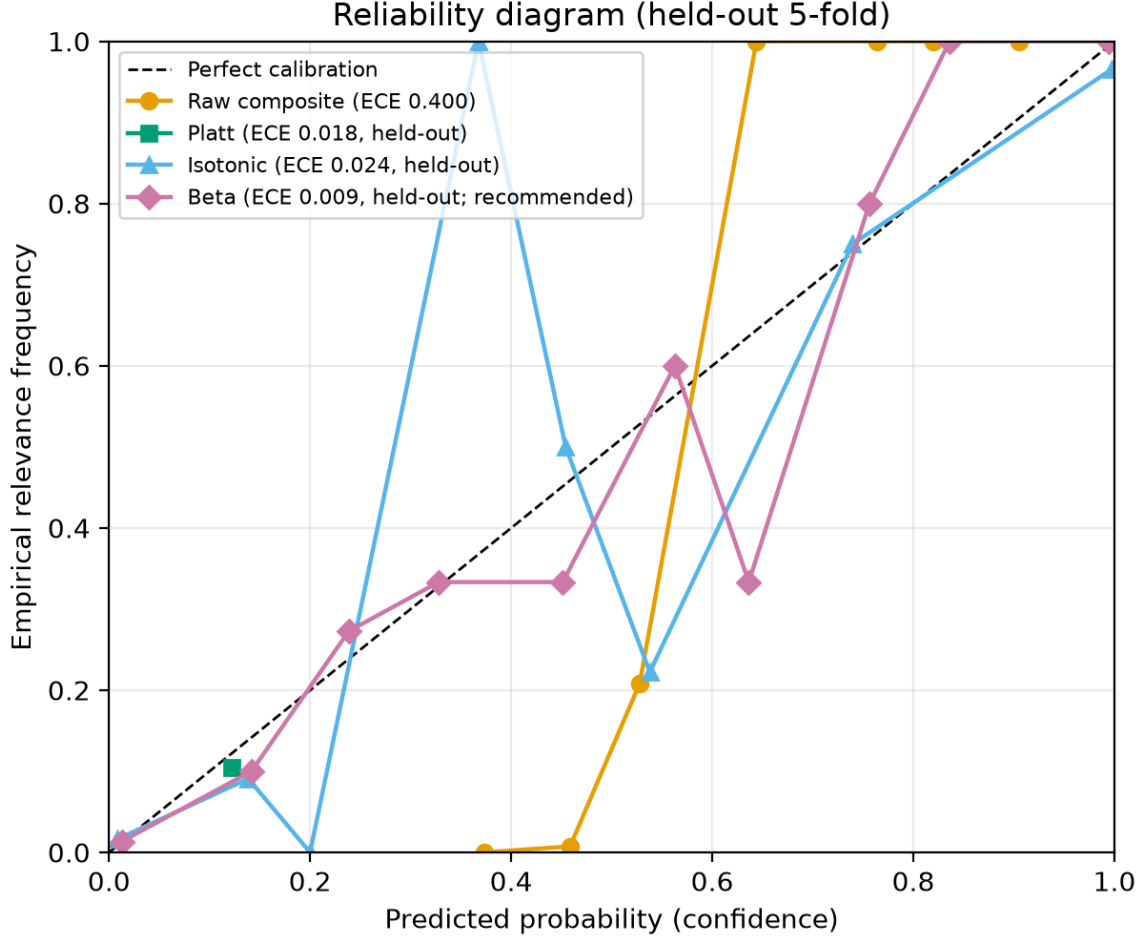


Figure 13: Reliability diagram (held-out 5-fold), regenerated from the calibration artifact `calibration_methods.json` (EXP-026). The raw composite (orange) is over-confident (ECE 0.40); Platt scaling (green) attains a low equal-width ECE (0.018) but compresses the confidence toward the base rate; isotonic (blue) tracks the diagonal at comparable ECE (0.024) while preserving discrimination; and beta calibration (magenta, the recommended map) tracks the diagonal at the lowest ECE (0.009) while preserving discrimination (Section 5.4, Table 5). Bins are 10 equal-width confidence intervals; $ECE = \sum_b |acc_b - conf_b| n_b / N$. The per-bin counts and the plot script are in the released artifact.